Identifiability-Aware Source Apportionment in City-Scale Advection-Diffusion Systems

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Abstract

Source apportionment from sparse urban air-quality sensors is an inverse problem whose resolution is limited by sensor placement, wind-driven transport, background variation, and noise. Known or proxy emission inventories make attribution scientifically meaningful by restricting the unknown source field to a finite set of candidate source groups, but they do not guarantee that those groups are distinguishable from the available observations. We represent time-varying source activity with a low-dimensional nonnegative temporal basis and formulate inventory-based source apportionment as a wind-conditioned lagged inverse problem in which each source–basis coefficient produces a sensor-time fingerprint. After projecting out a separate low-dimensional background space, the relevant object is the projected lagged source–basis response matrix ($\widetilde{H}_\Phi$). We show that exact identifiability at the chosen temporal-basis resolution requires full column rank of $\widetilde{H}_\Phi$, while noise-robust attribution is controlled by its singular values, coefficient visibility, background absorption, pairwise fingerprint coherence, and ray distance. Building on this analysis, we propose an identifiability-aware apportionment (IASA) framework that estimates nonnegative source–basis coefficients, reconstructs source activity trajectories, and reports uncertainty and conservative deterministic grouping recommendations for indistinguishable sources. We instantiate the framework on a New Delhi platform built from government $PM_{2.5}$ and wind observations, regulatory sensor locations, and four proxy source groups with declared temporal activity bases, and define controlled and observed-data evaluations of recovery, ambiguity, wind diversity, background stress, transport error, inventory robustness, and residual model adequacy. The framework reports the attribution resolution defensible conditional on the declared inventories, transport, background, lag, and noise model rather than presenting the most detailed possible attribution vector.

1 Introduction

Urban air pollution is driven by local emissions, meteorological transport, regional background, and sensor noise. For policy the central question is often not only *where* pollution is high but *which* source groups (traffic, industry, brick kilns, population-related activity) produced the observed concentrations, estimated from source inventories, transport models, and sparse measurements.

This is especially hard in sparse sensor networks: a model may fit sensor trajectories accurately yet fail to identify the underlying contributions. The reason is geometric. After wind transport and sparse observation, two distinct source groups can induce nearly identical sensor-time signatures, so a strong distant source and a weaker nearby source become observationally equivalent once attenuation, travel time, background correction, and noise are accounted for. Fine-grained attribution is then not merely uncertain; it is not identifiable from the sensing system.

We study this in the inventory-based setting. Rather than recovering an arbitrary source field, we assume a finite set of known or proxy source maps, which restrict attribution to a low-dimensional, interpretable model, and we represent each source's time variation as a nonnegative combination of a few temporal basis functions. The unknowns are then source–basis coefficients, from which activity trajectories and contributions are reconstructed. These restrictions alone do not make the problem identifiable: coefficient fingerprints can still be weakly visible, absorbed by background components, or mutually coherent after transport.

The key object is a projected lagged source-response matrix. For a fixed wind realization, transport operator, inventory, sensor layout, and lag window, each source–basis pair induces a finite-horizon sensor-space fingerprint; stacking them gives H_Φ^{lag}. After projecting out a low-dimensional background basis Q, the inverse problem is governed by $\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}}$, whose rank, singular values, coefficient visibility, background absorption, pairwise coherence, and ray distance determine which contributions can be separated and which should be reported only after merging or qualification. The goal is therefore not to return the most detailed contribution vector the inventory permits, but a conservative resolution defensible under the available sensors, wind, background, and noise, conditional on the declared source model, with source estimates reported alongside identifiability diagnostics and ambiguity certificates.

We make the following contributions:

- We formulate sparse-sensor, inventory-based source apportionment as a lagged wind-conditioned inverse problem over nonnegative source–temporal-basis coefficients, and identify the projected lagged response matrix $\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}}$ as the central object governing coefficient separability after transport, sparse observation, lag, and background correction.

- We derive diagnostics for exact and noise-robust identifiability, including rank, smallest singular value, condition number, effective rank, coefficient visibility, background absorption, pairwise fingerprint coherence, and ray distance.
- We propose an identifiability-aware apportionment (IASA) procedure that fits nonnegative source–basis coefficients, reconstructs source activity trajectories, reports uncertainty, flags weakly visible coefficients, recommends connected report groups when eligible fingerprints are indistinguishable, and adds a per-sensor spatial attribution that back-traces the transport response to upwind regions and source groups while inheriting the global identifiable resolution rather than asserting finer per-sensor separations.
- We instantiate the framework on a New Delhi platform (government $PM_{2.5}$ and wind records, regulatory sensor locations, kernel coordinate-query gridded wind imputation, and four normalized proxy source groups with declared temporal activity bases) and define controlled and observed-data evaluations of conditioning, source ambiguity, background stress, wind diversity, temporal-basis recovery, response error, uncertainty, and residual structure.

2 Related Work

Source apportionment has a long history in air-quality science, spanning receptor-oriented methods (chemical mass balance and factor-analytic approaches such as positive matrix factorization (Watson et al. 2002; Paatero and Tapper 1994; Hopke 2016)) and source-oriented models that propagate emissions inventories through transport, with reviews and harmonization efforts documenting their practical sensitivity to source profiles, tracer collinearity, factor interpretation, and inventory assumptions (Viana et al. 2008; Belis et al. 2013, 2019; Mircea et al. 2020; Hopke et al. 2020); this ambiguity is acute in settings such as India, where the same city yields divergent source-contribution estimates and policy-relevant categories are reported heterogeneously (Pant and Harrison 2012; Karagulian et al. 2015). Estimating latent quantities from partial observations is the province of inverse problems and data assimilation (Tarantola 2005; Engl, Hanke, and Neubauer 1996; Wang et al. 2023; Carrassi et al. 2018) and, more recently, physics-informed and operator-learning methods (Raissi, Perdikaris, and Karniadakis 2019; Li et al. 2024, 2020; Bhardwaj, Balashankar, and Subramanian 2025), which motivate our transport-structured, PyTorch-based constrained backend but do not by themselves resolve identifiability in a many-to-one sparse-sensing map. Graph, transformer, and sequence models address the same sparse-sensor regime (Li et al. 2017; Yu, Yin, and Zhu 2017; Wu et al. 2019; Chen et al. 2023; Iyer et al. 2022; Bhardwaj et al. 2025), but target forecasting, interpolation, or representation learning rather than attribution to physically meaningful inventories. Finally, our separability question is a form of the observability studied in control theory and system identification (Chen and Chen 1984; Ljung et al. 1987; Banks and Kunisch 2012; Colton et al. 1990). Unlike this prior work, we analyze when declared or proxy source groups are *distinguishable* from sparse concentration sensors after wind-conditioned transport, finite lags, background correction, and noise, and report the attribution resolution the sensing system supports; Appendix H expands each strand.

3 Problem Setup: Sparse Sensors, Wind, and Source Apportionment

We formalize source apportionment as a finite-dimensional inverse problem in which known source inventories are transported to a sparse sensor network under a wind-conditioned lagged operator, defining the observation model, inventory parameterization, background correction, and projected response matrix used throughout.

City-scale transport model. Let $u(\mathbf{x}, t) \in \mathbb{R}$ denote the pollutant concentration field on $\Omega \subset \mathbb{R}^d$. A generic continuous-time model is $\partial u/\partial t = \mathcal{F}_\eta(u, \mathbf{x}, t; w(t)) + s(\mathbf{x}, t) + b(\mathbf{x}, t)$, where $w(t)$ is the meteorological state, s is local source emission, b is background or regional pollution, and η denotes transport and dispersion parameters. The exact dynamics may be approximated by a plume model, an advection–diffusion solver, or a learned surrogate constrained by transport structure.

Sparse observations. Discretize the concentration field on n grid cells, so that $\mathbf{u}_t \in \mathbb{R}^n$. Let $X_{\mathrm{sens}} = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ be the fixed sensor locations with $m \ll n$. The observation operator $O : \mathbb{R}^n \to \mathbb{R}^m$ samples or interpolates the latent field at these locations, $\mathbf{y}_t = O\mathbf{u}_t + \boldsymbol{\epsilon}_t$ with $\mathbf{y}_t \in \mathbb{R}^m$; stacking over $t = 1, \ldots, T$ gives $Y = [\mathbf{y}_1^\top, \ldots, \mathbf{y}_T^\top]^\top \in \mathbb{R}^{mT}$. Because m is small relative to the field dimension, many latent fields and source configurations can be consistent with the same sensor trajectory.

Inventory-based source parameterization. We assume a set of K known or proxy source maps, $S = [\mathbf{s}_1\ \mathbf{s}_2\ \cdots\ \mathbf{s}_K] \in \mathbb{R}^{q\times K}$, where $\mathbf{s}_k \in \mathbb{R}^q$ is the spatial pattern of source group k. The source grid dimension q may differ from the concentration-grid dimension n. We represent time-varying source activity with a fixed, predeclared nonnegative temporal dictionary $\Phi = [\boldsymbol{\phi}_1\ \cdots\ \boldsymbol{\phi}_B] \in \mathbb{R}_+^{T\times B}$ and coefficients $C = [c_{kb}] \in \mathbb{R}_+^{K\times B}$. The activity of source k at time t is $\theta_k(t) = \sum_{b=1}^B c_{kb}\phi_b(t) \geq 0$. The basis Φ is shared as a dictionary of admissible temporal patterns, not as a claim that all sources follow the same activity trajectory: each source has its own nonnegative mixture weights c_{kb}. In applications, these basis functions may encode predeclared calendar, diurnal, seasonal, regulatory, or inventory-derived activity patterns, and source-specific prior knowledge can be imposed by masking inadmissible source–basis coefficients. We do not learn Φ from the same sparse concentration observations used for apportionment; jointly estimating both Φ and C would introduce additional bilinear non-identifiability beyond the source distinguishability analyzed here. The activity bases used for source emissions are nonnegative; signed temporal harmonics, when used, belong to the separate background basis Q. We vectorize C in source-major, basis-minor order as $\mathbf{c} = \mathrm{vec}_{\mathrm{src}}(C) \in \mathbb{R}_+^J$, where $J = KB$. The constant source activity model is the special case $B = 1$, $\phi_1(t) = 1$, and $c_{k1} = \theta_k$.

Lagged wind-conditioned response. Emissions released at time $t-\ell$ may affect sensors at time t only after being transported by the intervening wind field. Let $G^{\partial}_{t,t-\ell}(w_{t-\ell:t}) \in \mathbb{R}^{n\times q}$ map emissions released on the source grid at time $t-\ell$ to the concentration grid at time t. The superscript ∂ denotes an open-boundary operator: mass transported outside Ω leaves the modeled domain rather than being reflected or wrapped around. For a maximum lag L, define $\mathcal{L}_t = \{0, 1, \ldots, \min(L, t-1)\}$. For source k and basis component b, the corresponding unit-coefficient sensor fingerprint at time t is $\mathbf{h}^{\mathrm{lag}}_{t,kb} = \sum_{\ell\in\mathcal{L}_t} \phi_b(t-\ell)\, OG^{\partial}_{t,t-\ell}(w_{t-\ell:t})\mathbf{s}_k \in \mathbb{R}^m$. Stacking over time and arranging columns in source-major, basis-minor order gives $H^{\mathrm{lag}}_{\Phi} = [\mathbf{h}^{\mathrm{lag}}_{11} \ \cdots \ \mathbf{h}^{\mathrm{lag}}_{1B} \ \cdots \ \mathbf{h}^{\mathrm{lag}}_{KB}] \in \mathbb{R}^{mT\times J}$, $J = KB$, where each $\mathbf{h}^{\mathrm{lag}}_{kb} \in \mathbb{R}^{mT}$ is the finite-horizon unit-coefficient sensor fingerprint, the sensor-time pattern produced by one unit of coefficient c_{kb}. When $B = 1$, these columns reduce to the fingerprints of the constant-activity model.

The lag window changes these fingerprints and the release times contributing to each observation; it does not add sensor–time observations or change the row count. We choose it before fitting source coefficients. Physical travel-time and residence-time bounds define an increasing candidate grid $\mathcal{L}_{\mathrm{cand}}$. For adjacent candidates L and $L+\Delta$, with the same observed rows and columns, define

$$\eta_L = \frac{\|H^{\mathrm{lag}}_{\Phi}(L+\Delta) - H^{\mathrm{lag}}_{\Phi}(L)\|_F}{\max\{\|H^{\mathrm{lag}}_{\Phi}(L+\Delta)\|_F, \epsilon\}}, \tag{1}$$

where the subscript F is the Frobenius norm $\|A\|_F = (\sum_{i,j} A_{ij}^2)^{1/2}$. The primary lag is the smallest candidate with $\eta_L \le \tau_L$, using $\tau_L = 10^{-3}$ by default. Rank, conditioning, and ambiguity components are also reported across the candidate grid. The fitted value of $\mathbf{c}$ is never used to select L.

Background and temporal correction. Observed concentrations include components not explained by the local source inventory, including regional background, smooth temporal trends, and sensor-specific offsets. We represent these components with a low-dimensional basis $Q \in \mathbb{R}^{mT\times r}$, $\boldsymbol{\gamma} \in \mathbb{R}^r$, and write

$$Y = H^{\mathrm{lag}}_{\Phi}\mathbf{c} + Q\boldsymbol{\gamma} + E. \tag{2}$$

The low-dimensionality ensures that source apportionment is not confounded by background correction.

Real concentration records need not be complete. Let $\mathcal{O}$ be the ordered set of observed sensor–time $\mathrm{PM}_{2.5}$ rows and let $M_{\mathcal{O}} \in \{0,1\}^{N\times mT}$, $N = |\mathcal{O}|$, select them. The observation mask is applied identically to every row-aligned object:

$$Y_{\mathcal{O}} = M_{\mathcal{O}}Y, \quad H^{\mathrm{lag}}_{\Phi,\mathcal{O}} = M_{\mathcal{O}}H^{\mathrm{lag}}_{\Phi}, \quad Q_{\mathcal{O}} = M_{\mathcal{O}}Q. \tag{3}$$

Row metadata are filtered by the same ordered index set; an alignment mismatch is an error rather than an invitation to reorder silently. Wind may be imputed, but missing $\mathrm{PM}_{2.5}$ values are not. For complete controlled observations, $M_{\mathcal{O}} = I$ and $N = mT$. Below we drop the $\mathcal{O}$ subscript and use $Y \in \mathbb{R}^N$, $H^{\mathrm{lag}}_{\Phi} \in \mathbb{R}^{N\times J}$, and $Q \in \mathbb{R}^{N\times r}$ for these aligned arrays.

Let $P_Q^{\perp} = I - QQ^{\dagger}$ be the projection onto the orthogonal complement of the background space, with $Q^{\dagger}$ the Moore–Penrose pseudoinverse. The corrected inverse problem is

$$\widetilde{Y} = \widetilde{H}_{\Phi}\mathbf{c} + \widetilde{E}, \qquad \widetilde{Y} = P_Q^{\perp}Y, \qquad \widetilde{H}_{\Phi} = P_Q^{\perp}H^{\mathrm{lag}}_{\Phi}. \tag{4}$$

All identifiability diagnostics in this paper are computed from $\widetilde{H}_{\Phi}$, not from the raw inventory matrix or the unprojected response.

Projected source–basis fingerprints. Writing $H^{\mathrm{lag}}_{\Phi} = [\mathbf{h}^{\mathrm{lag}}_{11} \ \cdots \ \mathbf{h}^{\mathrm{lag}}_{KB}]$ and $\widetilde{H}_{\Phi} = [\widetilde{\mathbf{h}}_{11} \ \cdots \ \widetilde{\mathbf{h}}_{KB}]$, we have $\widetilde{\mathbf{h}}_{kb} = P_Q^{\perp}\mathbf{h}^{\mathrm{lag}}_{kb}$. These projected fingerprints encode the observable effect of each temporal component of each source after wind transport, lag, sparse sensing, and background correction. Source apportionment at the chosen temporal-basis resolution is well posed only to the extent that these coefficient fingerprints are visible and separable.

4 Wind-Conditioned Response

Building on the projected inverse problem of Section 3, we describe how the response matrix is built and how source activities are estimated; construction, background, and refinement details are deferred to Appendix I.

Masked wind preparation. Government wind direction is meteorological (it records where wind comes from), whereas transport needs the direction in which pollution moves. We convert direction WD and speed WS to transport vectors

$$U_x = -WS\sin(WD\pi/180), V_y = -WS\cos(WD\pi/180), \tag{5}$$

supply the station vectors and masks to a coordinate-query wind imputer, and query it on every response-grid cell and hour to obtain a gridded transport field $\widehat{\mathbf{w}}_t(\mathbf{x}_g)$, converted to grid displacement before advection. The adopted imputer is a normalized Gaussian-kernel interpolator over the station vectors; a learned coordinate-query model was trained but not adopted, as it did not beat a city-mean baseline on held-out station–time error (Section 7.4). The gridded field is an estimated input, not ground truth: to propagate wind uncertainty we build transport ensembles kept separate from inventory-robustness scenarios. Controlled experiments use the same interface for constant, single-direction, diurnal, AR(1), and multi-directional synthetic winds. The imputer query, unit conversion, and ensemble construction are detailed in Appendix I.

Open-boundary lagged plume response. The transport response $G^{\partial}_{t,\tau}$ maps emissions released at $\tau \le t$ to the concentration field at time t. It is realized by an open-boundary Gaussian puff approximation: each source cell releases a puff whose center is advected by the interpolated wind field and spread by an anisotropic Gaussian kernel, evaluated directly at sensor coordinates. The operator preserves nonnegativity and is mass non-increasing inside the modeled domain ($G^{\partial}_{t,\tau}\mathbf{z} \ge 0$ and $\mathbf{1}^{\top}G^{\partial}_{t,\tau}\mathbf{z} \le \mathbf{1}^{\top}\mathbf{z}$ for $\mathbf{z} \ge 0$); when a puff center exits Ω the whole remaining release is removed and never reflected, wrapped, clamped, or reinserted. Boundary, truncation, and whole-release exit losses are recorded only as diagnostics and never used to renormalize the response, and because the kernel is evaluated at sensor points rather than integrated over grid cells, summing entries across sensors or times does not estimate retained mass. Response matrices are built for source-induced increments relative to a declared, pre-fit initial-condition baseline. The puff dynamics, dispersion covariance, exit bookkeeping, baseline policy, and PyTorch differentiability are given in Appendix I.

Background construction and projection. The primary background basis Q is specified before source fitting from source-independent metadata only: timestamps, day labels, sensor identity, and sensor coordinates (a global constant, elapsed-time polynomials, daily sine/cosine harmonics, day and sensor offsets, and coordinate trends), with effective rank capped at eight; for the New Delhi study it is the rank-four member (constant, centered linear trend, first daily sine and co-sine). Columns built from inventories, response fingerprints, or fitted source signals are excluded and permitted only in labeled stress tests that measure how source-like background variation can absorb attribution signal. Once Q is fixed, the projection $P_Q^{\perp}$ of Section 3 is applied implicitly through a thin SVD of Q, requiring identical row metadata across Y, H_Φ^{lag}, and Q. Alternative source-independent bases are pre-declared sensitivity analyses, not data-selected replacements (Appendix I).

Response matrix and source activity fit. Given the wind field, inventory, and temporal basis, the lagged unit-coefficient response has (k, b)-th column

$$\mathbf{h}_{t,kb}^{\text{lag}} = \sum_{\ell \in \mathcal{L}_t} \phi_b(t-\ell) O G_{t,t-\ell}^{\partial}(\widehat{\mathbf{w}}_{t-\ell:t}; \psi)\mathbf{s}_k, \tag{6}$$

and stacking over time in source-major, basis-minor order yields $H_\Phi^{\text{lag}} \in \mathbb{R}^{N\times J}$, $J = KB$, after the observation selection (Eq. 3); projection then gives $\widetilde{Y}, \widetilde{H}_\Phi$ (Eq. 4). Scientific knowledge may declare a set $\mathcal{F}_0$ of coefficients identically zero before the response is diagnosed or fitted; with $\mathcal{I} = \{1, \dots, J\} \setminus \mathcal{F}_0$, IASA removes those columns, computes every diagnostic and the fit on $\widetilde{H}_{\Phi,\mathcal{I}}$, and restores zeros afterward (default $\mathcal{F}_0 = \varnothing$). Coefficients are never deleted after fitting merely because their estimates are small, which would change the diagnosed inverse problem after seeing the data and could make the reported identifiability artificially favorable. Source–basis coefficients are estimated by nonnegative regularized least squares

$$\widehat{\mathbf{c}} = \arg\min_{\mathbf{c}\geq 0} \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda R(\mathbf{c}), \tag{7}$$

with R a ridge or weak inventory-prior penalty, and reshaped into activity trajectories

$$\widehat{\theta}_k(t) = \sum_{b=1}^{B} \widehat{c}_{kb}\phi_b(t). \tag{8}$$

The projected formulation is used for identifiability because it exposes the part of each fingerprint that remains after background correction; the regularizer choices, an equivalent joint source/background fit, and an optional constrained end-to-end refinement of wind, dispersion, and coefficients, accepted only if it does not degrade identifiability, are given in Appendix I.

5 Source Apportionment Identifiability

We now characterize when source activity at a chosen temporal-basis resolution is determined by the projected observations. Throughout this section $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}$ with $\widetilde{H}_\Phi = P_Q^{\perp} H_\Phi^{\text{lag}} \in \mathbb{R}^{N\times J}$ and $J = KB$. The analysis is conditional on the constructed wind field, transport operator, source inventories, temporal basis, lag window, and background basis. We use the flattened coefficient index $j = (k, b)$; "source fingerprint" without a basis qualifier refers only to the constant-activity case $B = 1$.

5.1 Exact Identifiability

Definition 5.1 (Exact identifiability)**.** The source–basis coefficient vector is identifiable from $\widetilde{Y}$ if $\widetilde{H}_\Phi \mathbf{c}_1 = \widetilde{H}_\Phi \mathbf{c}_2 \Longrightarrow \mathbf{c}_1 = \mathbf{c}_2$. This is identifiability of the reconstructed source activity trajectories at the selected temporal-basis resolution.

Here $\mathbb{R}_+^J = \{\mathbf{c} \in \mathbb{R}^J : c_j \geq 0 \text{ for all } j\}$ is the nonnegative orthant, i.e. the feasible set of nonnegative source–basis coefficient vectors.

Proposition 5.2 (Identifiability of inventory-based apportionment)**.** *In the noiseless projected model $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c}$, $\mathbf{c}$ is identifiable uniformly for all $\mathbf{c} \in \mathbb{R}_+^J$ if and only if*

$$\operatorname{rank}(\widetilde{H}_\Phi) = J. \tag{9}$$

The proof (rank-nullity in both directions, with a nonnegative-feasible counterexample when the rank is deficient) is given in Appendix B. We use this full-column-rank condition as our identifiability criterion for the declared source–basis coefficient set. Coefficients fixed to zero by scientific knowledge are masked before fitting; the theorem and all diagnostics then apply to the reduced, predeclared column set. A near-zero fitted estimate does not justify deleting its column after fitting or recomputing diagnostics on a more favorable response matrix. Rank deficiency may arise from spatial source ambiguity, dependence among temporal modes, or their interaction.

5.2 Noise-Robust Identifiability

Let $\sigma_1(\widetilde{H}_\Phi) \geq \cdots \geq \sigma_J(\widetilde{H}_\Phi) \geq 0$ be the square roots of the eigenvalues of $\widetilde{H}_\Phi^\top \widetilde{H}_\Phi$, including zeros when $J > N$ or the matrix is rank deficient. Even with full rank, small singular values amplify noise.

Proposition 5.3 (Noise-robust apportionment)**.** *Assume $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}$ and $\operatorname{rank}(\widetilde{H}_\Phi) = J$. Let $\widehat{\mathbf{c}}_{\text{LS}}$ be the unconstrained least-squares estimate. Then $\|\widehat{\mathbf{c}}_{\text{LS}} - \mathbf{c}\|_2 \leq \|\widetilde{E}\|_2/\sigma_J(\widetilde{H}_\Phi)$. If $\widehat{\mathbf{c}}$ is the nonnegative least-squares estimate and $\mathbf{c} \geq 0$, then*

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 \leq \frac{2\|\widetilde{E}\|_2}{\sigma_J(\widetilde{H}_\Phi)}. \tag{10}$$

Both bounds follow from the pseudoinverse identity $\|\widetilde{H}_\Phi^\dagger\|_2 = 1/\sigma_J(\widetilde{H}_\Phi)$ together with feasibility of the true $\mathbf{c}$ in the nonnegative program; the derivation is in Appendix B. Thus $\sigma_J(\widetilde{H}_\Phi)$ measures worst-case robustness: source apportionment can be exactly identifiable but practically unstable when this quantity is small.

5.3 Identifiability Diagnostics

For floating-point computation, the default numerical rank tolerance is $\tau_{\mathrm{num}} = \max(N, J)\,\epsilon_{\mathrm{mach}}\,\sigma_1$ with numerical rank $r_{\mathrm{num}} = \#\{i : \sigma_i > \tau_{\mathrm{num}}\}$, where ϵ_{mach} is the machine precision of the SVD dtype (about 2.2×10^{-16} in double precision). This τ_{num} is a purely numerical tolerance for detecting exact rank deficiency; it is distinct from the scientific, noise-dependent threshold τ_σ introduced below. The primary identifiability score is the *padded minimum singular value* $\sigma_J(\widetilde{H}_\Phi)$, where the singular spectrum is padded with zeros whenever $J > N$ or $\widetilde{H}_\Phi$ is rank deficient. A rank-deficient response therefore scores zero and is never reported as stable. The condition number is $\kappa(\widetilde{H}_\Phi) = \sigma_1/\sigma_J$ when $r_{\mathrm{num}} = J$ and ∞ when $r_{\mathrm{num}} < J$. The smallest numerically positive value may additionally be reported as $\sigma_{\min,+}$, but it is not the identifiability score and never turns a rank-deficient matrix into a stable one. For a separately chosen, noise-dependent threshold τ_σ, the effective rank is $r_{\mathrm{eff}}(\tau_\sigma) = \#\{i : \sigma_i(\widetilde{H}_\Phi) > \tau_\sigma\}$. Unlike τ_{num}, the threshold τ_σ is set from the observation-noise scale, not from machine precision. Given a per-row noise standard deviation σ_e (from sensor calibration or a bootstrap residual estimate), a singular direction is only resolvable when its singular value exceeds the level at which iid noise projects onto it; a workable predeclared default is $\tau_\sigma = \sigma_e\sqrt{N}$. It is recorded independently of τ_{num} and is not tuned on source-recovery results. If $r_{\mathrm{eff}}(\tau_\sigma) < J$, the sensor network does not support reliable estimation of all J source–basis coefficients at that noise level.

For coefficient-specific diagnostics, let $\widetilde{\mathbf{h}}_j$, $j = (k, b)$, be a projected source–basis fingerprint. Its visibility and the weak set at threshold $\tau_v \geq 0$ are $v_j = \|\widetilde{\mathbf{h}}_j\|_2$ and $\mathcal{W} = \{j : v_j \leq \tau_v\}$. A weak coefficient has little measurable effect after transport and background correction: v_j is the sensor-time magnitude produced by one unit of coefficient c_j, and τ_v marks fingerprints below the minimum detectable signal. A practical predeclared rule sets τ_v from the noise scale and a target signal-to-noise ratio, e.g. $\tau_v = \mathrm{SNR}_{\min}\,\sigma_e$, so a unit coefficient is flagged weak when even its full response is indistinguishable from noise. For eligible indices $i, j \notin \mathcal{W}$, pairwise ambiguity is measured by projected fingerprint coherence $\rho_{ij} = |\widetilde{\mathbf{h}}_i^\top \widetilde{\mathbf{h}}_j|/(\|\widetilde{\mathbf{h}}_i\|_2\|\widetilde{\mathbf{h}}_j\|_2)$. Values near one indicate nearly proportional sensor-time signatures. The ambiguous-pair set is $\mathcal{A} = \{(i, j) : i, j \notin \mathcal{W},\ \rho_{ij} > \tau_\rho\}$, $\tau_\rho \in [0, 1]$. The threshold τ_ρ is predeclared or calibrated on controlled experiments, not optimized after inspecting source recovery; a conservative default takes τ_ρ close to one so that only near-proportional fingerprints are merged. Coherence and ray distance are reported only for eligible (nonweak) pairs; entries involving a weak fingerprint are shown as "N/A" and excluded from extrema, while the weak-visibility flag is always reported.

To express the same scale-invariant geometry as a distance, we also report the ray distance for eligible, generally signed projected fingerprints, $d_{ij}^{\mathrm{ray}} = \min_{\alpha\in\mathbb{R}} \|\widetilde{\mathbf{h}}_i - \alpha\widetilde{\mathbf{h}}_j\|_2/\|\widetilde{\mathbf{h}}_i\|_2$. For nonweak fingerprints,

$$d_{ij}^{\mathrm{ray}} = \sqrt{1 - \rho_{ij}^2}, \tag{11}$$

which we derive in Appendix B; numerical implementations evaluate $\sqrt{\max(0, 1 - \rho_{ij}^2)}$. This is an unoriented linear-ray diagnostic, not a claim that projected fingerprints are nonnegative. High coherence and small ray distance encode the same pairwise geometry, so ray distance is a diagnostic geometrically equivalent to coherence under this normalization, not an independent identifiability criterion. Like coherence, it is "N/A" for pairs involving $\mathcal{W}$ and excluded from distance extrema.

Background absorption. Background correction can improve robustness by removing broad temporal variation, but it can also remove source-driven signal. Let $P_Q = QQ^\dagger$. For coefficient fingerprint $j = (k, b)$, define the background absorption fraction $a_j = \|P_Q\mathbf{h}_j^{\mathrm{lag}}\|_2/\|\mathbf{h}_j^{\mathrm{lag}}\|_2$. If $\|\mathbf{h}_j^{\mathrm{lag}}\|_2$ is zero at numerical tolerance, a_j is undefined and reported as N/A; the coefficient remains explicitly weak. If $a_j \approx 1$, most of the raw source–basis fingerprint lies in the background space, and the projected visibility v_j will be small even if the raw fingerprint is large. This is why identifiability must be assessed after projection.

Response-matrix perturbations. The constructed response can be wrong for two scientifically different reasons: transport error (wind interpolation, transport approximation, dispersion) and inventory error (missing, misplaced, misclassified, or incorrectly scaled sources). Writing the projected discrepancy as $\Delta_H = \Delta_{\mathrm{tr}} + \Delta_{\mathrm{inv}} + \Delta_{\mathrm{int}}$, response error acts like additional observation error: a perturbation $\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \leq \delta_H$ inflates the least-squares error bound to $(\|\widetilde{E}\|_2 + \delta_H\|\mathbf{c}\|_2)/\sigma_J(\widehat{\widetilde{H}}_\Phi)$ (Proposition and proof in Appendix J), so either error is most damaging when the projected response is already ill-conditioned. Crucially, the transport term Δ_{tr} may be propagated through ensembles into uncertainty intervals, whereas the inventory term Δ_{inv} is reported only as named robustness scenarios that are never pooled into an interval.

5.4 Identifiable Source Resolution

When coefficient fingerprints are weak or highly coherent, separate source estimates can be misleading. We define the source ambiguity graph by $\mathcal{E}_{\mathrm{src}} = \{(k, k') : \exists b, b'\ ((k, b), (k', b')) \in \mathcal{A}\}$. Each edge retains the maximum eligible coefficient coherence, minimum ray distance, and the source–basis pair attaining the trigger. Deterministically ordered connected components $\mathcal{G} = \{G_1, \ldots, G_M\}$ are recommended report groups. Weak coefficients remain separate flags and do not create edges. Rank deficiency with no eligible edge produces a global unresolved warning rather than an invented merge.

These components are deterministic, conservative reporting units, not a claim of the globally finest identifiable partition. In particular, if edges join A to B and B to C, the component contains $\{A, B, C\}$ even when A and C are distinguishable as a pair. The edge table therefore retains both triggering pairs and their diagnostic values so that this transitive over-merging is visible.

The coefficient fit is not silently replaced by a grouped refit. For reporting, group activity and fitted sensor contribution are the sums $\widehat{\theta}_{G_a}(t) = \sum_{k \in G_a} \widehat{\theta}_k(t)$ and $\widehat{Y}_{G_a} = \sum_{k \in G_a} \sum_b \mathbf{h}_{kb}^{\mathrm{lag}} \widehat{c}_{kb}$. Matrix diagnostics remain coefficient level; these group summaries express the coarser claims supported by the fit without inventing a scalar source fingerprint. Table 2 in Appendix A summarizes the scalar diagnostics and the set- and graph-valued reporting outputs.

Per-sensor source footprints and spatial attribution. The diagnostics above certify which coefficients are separable in the pooled sensor-time space; policy questions are often also spatial. Because the response is linear in $\mathbf{c}$, the fitted solution already localizes each monitor's signal without a new model: per-sensor contributions decompose exactly, and the sensor footprint is the pullback of the response row onto the source grid: the puff response of Section 4 read backward from the sensor, requiring no new operator (Appendix K). Identifiability is inherited, not created: for the row submatrix $\widetilde{H}_{\Phi}^{(s)}$ of a single sensor, $\sigma_J(\widetilde{H}_{\Phi}^{(s)}) \leq \sigma_J(\widetilde{H}_{\Phi})$ and $\mathrm{rank}(\widetilde{H}_{\Phi}^{(s)}) \leq \mathrm{rank}(\widetilde{H}_{\Phi})$, so a single sensor is never more identifiable than the pooled network. Footprints are therefore reported as spatial overlays of the global fit, with per-sensor source *shares* asserted only at the globally identifiable resolution, aggregating to a report group wherever the pooled diagnostics require one.

6 Identifiability-Aware Apportionment

We combine the estimator from Section 4 with the diagnostics from Section 5. The result is an identifiability-aware source apportionment procedure, abbreviated IASA, that estimates source contributions only at the resolution supported by the projected response matrix.

6.1 Inputs and Outputs

IASA takes sparse pollution observations and their row mask, sensor locations X_{sens}, wind observations and masks, source inventories $S \in \mathbb{R}^{q \times K}$, a background basis Q, a maximum lag L, a nonnegative temporal basis $\Phi \in \mathbb{R}_{+}^{T \times B}$, and an open-boundary transport response G^{∂}. It also uses a numerical SVD tolerance τ_{num} and distinct thresholds τ_σ, τ_ρ, and τ_v for effective rank, pairwise ambiguity, and visibility.

The outputs are source–basis coefficient estimates, reconstructed source activities and contribution estimates at the identifiable grouping, fitted trajectories and residuals, uncertainty intervals, singular-value, coherence, and ray-distance diagnostics, low-visibility flags, and merge recommendations.

The thresholds $\tau_{\mathrm{num}}, \tau_\sigma, \tau_v, \tau_\rho$ (with the refinement coherence cap τ_ρ^{ref} and wind-drift cap ϵ_w) are each predeclared or calibrated from noise or physics and none is tuned on

Algorithm 1 Identifiability-Aware Source Apportionment (IASA).

Input: observations Y and observed-row mask, sensors X_{sens}, wind observations and masks, inventories $S \in \mathbb{R}^{q \times K}$, background basis Q, maximum lag L, temporal basis $\Phi \in \mathbb{R}_{+}^{T \times B}$, transport response G^{∂}, thresholds $\tau_{\mathrm{num}}, \tau_\sigma, \tau_\rho, \tau_v$.

Output: source–basis coefficients and activities at the identifiable grouping, fitted trajectories/residuals, uncertainty intervals, diagnostics, low-visibility flags, and merge recommendations.

1: **Predeclare** the primary Q, the physical lag-candidate grid, Φ, the inventory version, and the fixed-zero mask $\mathcal{F}_0$, before inspecting any fit.

2: **Wind:** convert WD/WS to transport vectors (Eq. 5), preserving masks/observed values for audit; query the adopted kernel imputer on each response-grid cell for $\widehat{\mathbf{w}}_t(\mathbf{x}_g)$ and convert to grid displacement.

3: **Response & lag:** over unmasked source–basis pairs build $\mathbf{h}_{t,kb}^{\mathrm{lag}}$ (Eq. 6); select the smallest L with $\eta_L \leq \tau_L$ from geometry alone (Eq. 1), retaining the full lag sweep.

4: **Align & project:** apply the observed-$PM_{2.5}$ selection identically to Y, H, Q and metadata (Eq. 3) with source-major, basis-minor columns; form $\widetilde{Y}$, $\widetilde{H}_{\Phi}$ (Eq. 4).

5: **Fit:** solve $\widehat{\mathbf{c}}$ (Eq. 7) and reconstruct $\widehat{\theta}_k$ (Eq. 8).

6: **Diagnose:** compute r_{num}, the singular values, $r_{\mathrm{eff}}(\tau_\sigma)$, visibility v_j, absorption a_j, coherence ρ_{ij}, and ray distance d_{ij}^{ray}.

7: **Flag:** weak set $\mathcal{W} = \{j : v_j \leq \tau_v\}$; eligible ambiguous pairs $\mathcal{A} = \{(i,j) : i, j \notin \mathcal{W}, \rho_{ij} > \tau_\rho\}$.

8: **Group:** take deterministic connected components of the source ambiguity graph as conservative report groups (Sec. 5.4); weak coefficients are flagged but create no edges, and group activities/contributions are member sums that do not replace the coefficient fit.

9: **Adequacy:** if an externally calibrated noise model exists, run the refitted parametric-bootstrap residual test (Sec. 6.3); otherwise label the residual summary uncalibrated.

10: **Report** source estimates, observation/transport uncertainty, inventory robustness scenarios, residual adequacy, diagnostics, and conservative source groups.

source-recovery results; they are summarized in Appendix L (Table 3).

6.2 Procedure

Algorithm 1 states the procedure. Its step order enforces the method's central discipline of separating fit from diagnosis: all model choices, thresholds, and the fixed-zero mask are predeclared (line 1), the response is built, projected, and fitted (lines 2–5), and only afterwards are the identifiability diagnostics computed and conservative groups reported (lines 6–8). Because no threshold or column set is chosen after inspecting the fitted coefficients (the lag itself is selected from response-matrix geometry, not from $\widehat{\mathbf{c}}$), the reported identifiability cannot be made artificially favorable by post-fit tuning.

This procedure separates two questions: which source activity trajectories best fit the projected sensor data, and which source–basis coefficients are supported by the geometry of $\widetilde{H}_{\Phi}$.

The objective in Equation 7 is a smooth convex quadratic under a simple nonnegativity constraint, so we solve it with projected FISTA (Beck and Teboulle 2009): an accelerated proximal-gradient method whose proximal step for the indicator of $\{\mathbf{c} \geq 0\}$ is the clamp $\mathbf{c} \leftarrow \max(\mathbf{c}, 0)$. Each gradient step uses a Lipschitz step size derived from the largest singular value, with monotone restart, and convergence is checked on the projected-gradient/KKT residual and the relative objective change. Device, dtype, convergence status, iteration count, and the final KKT residual are part of the fit record. Transport-ensemble and bootstrap refits are batched on the same device when their shapes agree.

6.3 Model Adequacy, Uncertainty, and Reporting

IASA attaches uncertainty to reported contributions from an active-set/ridge covariance of the fitted coefficients and, when transport ensembles are supplied, from empirical quantiles across ensemble refits; transport uncertainty (`ensemble_kind=transport`) and inventory-robustness scenarios (`ensemble_kind=inventory`) are never pooled into one interval (Appendix L).

Residual magnitude alone is not calibrated evidence of misspecification. When an externally calibrated observation-noise model Σ_e is available, IASA computes a background-orthogonal residual statistic T_{res} whose null distribution (not a plain χ^2, since $\widehat{\mathbf{c}}$ is fitted from the same data) is calibrated by a parametric bootstrap ($B_{\mathrm{res}} = 1000$ refits, $\alpha = 0.05$; Appendix L). The test is one-sided: a rejection establishes a mismatch with the declared model but not its cause, whereas a non-rejection does not establish inventory completeness: an omitted source whose signature lies in $\operatorname{span}[H_\Phi^{\mathrm{lag}}, Q]$ is absorbed by fitted terms and stays residual-invisible. If no calibrated noise model is supplied, IASA reports raw and projected residual summaries with `calibration_status=uncalibrated` and emits no pass/fail claim. Prospective network adequacy over a distribution of historical or simulated wind windows, and acceptance criteria for the optional refinement, are described in Appendix L.

For each reported source group, IASA returns the group activity trajectory, its uncertainty interval, member visibilities, and the maximum eligible coherence with other groups; a contribution is marked reliable only when its coefficients are above the visibility threshold, its interval is sufficiently narrow, and its eligible coherence is below the ambiguity threshold, and is otherwise labeled weakly visible, ambiguous, or merged into a coarser group. It additionally reports the per-sensor contribution decomposition and source footprints of Section 5.4 as spatial overlays of the same global fit, aggregated to the identifiable report groups wherever the pooled diagnostics require grouping.

7 Evaluation

We evaluate on a single New Delhi platform so that the controlled and observed-data studies share one regulatory sensor geometry, one $PM_{2.5}$ and wind record, and one set of proxy inventories, and we ask whether the geometry of $\widetilde{H}_\Phi$ predicts which coefficient, activity, and grouped-source claims are defensible. A shared base configuration is varied along one controlled axis at a time, avoiding conflation of wind, source geometry, background flexibility, noise, and forward-model error; all runs use the 40×40 resolution, and unless stated otherwise the controlled experiments use a 72-hour transport window and the observed study four consecutive one-week windows. Real imputed wind is a controlled transport input when coefficients are synthetically known; the observed-$PM_{2.5}$ mode is a separate evaluation and is never assigned synthetic source-recovery metrics.

7.1 New Delhi Experimental Platform

We define the platform once; the studies that follow vary wind, noise, backgrounds, source ambiguity, activity bases, and forward-model error within it.

Government observations and sensor network. We use hourly government observations (CPCB 2023, 2025) (columns `monitor_id`, `timestamp_round`, `AT`, `RH`, `WD`, `WS`, `pm10`, `pm25`) spanning 21,960 hourly timestamps from 1 May 2018 through 31 October 2020 in Indian Standard Time; approximately 74.9% of wind-direction, 72.8% of wind-speed, and 90.5% of $PM_{2.5}$ entries are observed, and a valid-value mask (`True` where the record is present) is retained alongside station coordinates and timestamps. The two Pusa monitors are not treated as coincident independent sensors: `Pusa_IMD` and `Pusa_DPCC` are averaged by timestamp into `Pusa_averaged` (mean coordinates, per-variable hourly averaging), giving the final 32-sensor layout. $PM_{2.5}$ is never imputed: its flattened valid-value mask defines M_O in Equation 3, and the same ordered sensor–time rows are retained in Y, H_Φ^{lag}, Q, and their metadata.

Wind preparation. The observed WD/WS values are converted using Equation 5 and completed on the response grid by the kernel coordinate-query imputer of Section 4; a learned imputer was trained and evaluated but not adopted (Section 7.4). The saved wind product retains raw station values, masks, station coordinates, held-out validation splits, checkpoint metadata, and the gridded query field used by the response operator.

Inventories and proxy temporal activities. The study uses four source groups (brick kilns, industries, population density, and traffic; Table 4 in Appendix M), each with one spatial proxy map and its own admissible set of temporal-basis components. Brick-kiln and industry proxies follow the regional emissions inventory (Guttikunda and Calori 2013), population density uses the gridded population product (Columbia University 2018), and the traffic map is derived from road map data (Google 2024). Every 80×80 map is cropped to the New Delhi study window (rows 21:61, columns 16:56) covering the regulatory network, giving a 40×40 map, then divided by its own cropped 99th percentile. Traffic is modeled as a single road-network source whose nearest-slot maps at 00:00, 06:00, 12:00, and 18:00 define traffic-specific temporal-basis components under the shared dictionary Φ and the fixed-zero mask $\mathcal{F}_0$, so one road map carries a time-varying activity (assuming the congestion *spatial* pattern is time-invariant and only its amplitude varies). Brick-kiln activity uses deterministic alternating 12-hour blocks, industry a continuous operating fraction with higher daytime (07:00–19:00) activity, and population a baseline with smooth cooking peaks; these are deterministic study assumptions, not observed emissions truth. Because each map uses independent proxy normalization, fitted coefficients and contributions are in normalized proxy units, not physical emission totals or shares; no learned free residual source map is used, and unexplained broad signal belongs to the background model or residual diagnostics.

Forward models, sensor observations, and backgrounds. Both forward paths use the New Delhi-derived 40×40 domain and regulatory sensor coordinates, with the open-boundary Gaussian puff operator of Section 4 evaluated directly at sensors. The same operator is the matched generator for controlled recovery experiments (Experiments 1–4) and the inference operator for observed $\mathrm{PM}_{2.5}$; only the initial-condition baseline differs by mode: a zero-source, zero-initial-state baseline for controlled operator-matched runs, and a subtracted zero-source transported kriged-initial-condition baseline for observed runs. An auxiliary edge-hold advection–diffusion simulator provides a *structural* forward-model mismatch (data generated by a distinct transport operator, fit with the puff response) that also drives the residual-adequacy test of Section 6.3 (Experiment 5). Normal backgrounds follow Section 4, with the base controlled background the rank-four member; the kriged-baseline construction and simulator settings are given in Appendix M.

Controlled and observed-data study modes. The controlled mode preserves New Delhi sensor geometry and transport while assigning known nonnegative source–basis coefficients, using synthetic or real gridded wind, controlled observation noise, source splits, sensor-layout variants, and optional operator mismatch. Where the study is explicitly about inventory geometry (Experiments 2 and 6) the real New Delhi inventory maps are used; the remaining controlled experiments use compact synthetic source cells for tractability, since their conclusions concern conditioning, coherence, and recovery rather than specific inventory footprints. Because its coefficients are known, it supports recovery and interval-coverage metrics. The observed-data mode combines real $\mathrm{PM}_{2.5}$, its observation mask, the gridded kernel-imputed wind field, the named normalized inventories, and the declared proxy temporal bases; it has no ground-truth source activities and is run over four consecutive one-week windows, each fitted and reported independently, reporting residuals, response geometry, uncertainty, weak and ambiguous coefficients, normalized proxy contributions, and recommended report groups rather than synthetic recovery error.

7.2 Metrics and Reporting

Controlled trials report absolute and relative coefficient error, reconstructed-activity error, per-source and grouped activity/contribution error, projected sensor residual, uncertainty-interval coverage, and correlations between diagnostic values and attribution error. All trials additionally report the padded singular spectrum, σ_J, numerical and effective rank, condition number, coefficient visibility, background absorption, eligible coherence and ray distance, weak flags, triggering coefficient pairs, and deterministic report groups (definitions from Section 5), accompanied by lag convergence, the fixed-zero mask and index mapping, primary background specification, and ensemble provenance, with transport uncertainty and inventory robustness in separate fields. Observed New Delhi runs instead report raw and projected residuals, fitted trajectories, normalized proxy coefficients and activities, sensor-space fitted contributions, observation/transport uncertainty intervals, inventory-scenario robustness, weak/ambiguous flags, conservative grouped contributions, and the residual calibration status; a reported percentage is explicitly a fraction of the fitted inventory-attributed sensor signal, not a physical-emission share.

7.3 Controlled Results

The controlled trials have known source–basis coefficients and therefore recovery and coverage metrics. Figure 1 consolidates the three experiments that establish the central claim (response-matrix geometry, not noise alone, sets the recovery ceiling), and the adequacy study below tests residual adequacy. Six further controlled experiments (coherence and grouped reporting, transport error, inventory robustness, lag-window selection, temporal-basis recovery, and per-sensor footprints) are reported in Appendix N.

Conditioning and Recovery. We vary source geometry and Gaussian observation noise at 0%, 1%, 5%, 10%, and 20% of the maximum clean sensor signal. For each trial we compare coefficient and reconstructed-activity errors with σ_J, numerical and effective rank, condition number, and visibility. Figure 1(a) reports the sweep. The design is full rank in both geometries (numerical rank = effective rank = 2); the fixed geometry fixes σ_J and the condition number, and coefficient error then grows linearly with the injected noise. The better-conditioned *close* geometry ($\sigma_J = 9.60$, $\kappa = 2.26$) recovers proportionally more accurately than the *separated* geometry ($\sigma_J = 3.27$, $\kappa = 8.89$): at 20% noise the coefficient relative error is 0.11 versus 0.26. This is our central claim: response-matrix geometry, not noise alone, sets the recovery ceiling. (Reconstructed-activity error and interval coverage are reported for the temporal-basis and adequacy studies, Experiments 9 and 8; Experiment 1 isolates the coefficient-error/conditioning relationship.) The projected sensor residual grows linearly with the injected noise, through 0, 6.37, 12.73, 25.46 (separated) and 0, 8.42, 16.84, 33.68 (close) at $0/5/10/20\%$ noise.

Background Stress. We compare no background, the normal rank-four basis from Section 7.1, a redundant-column basis with the same span, and a labeled source-like stress basis. This tests background flexibility directly: too little background leaves broad confounding in the residual, while a too-flexible or source-like basis can absorb source-driven signal and reduce identifiability. The response geometry is audited through visibility and the removed/raw absorption ratio. The rank-four primary and every sensitivity variant are declared before fitting; no basis is selected from Y or from its source-recovery score. Figure 1(c) compares the four predeclared bases (all fixed before fitting: `declared_before_fit` is true throughout). The redundant basis is byte-identical to the

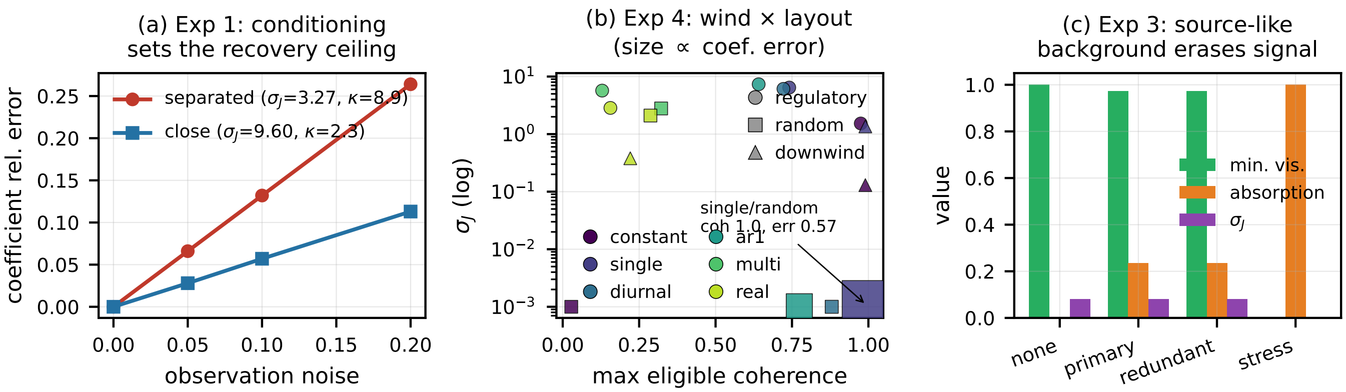

Figure 1: Controlled identifiability geometry governs recovery. **(a)** Experiment 1: coefficient relative error grows linearly with observation noise, and the better-conditioned *close* geometry ($\sigma_J = 9.60$, $\kappa = 2.26$) recovers proportionally more accurately than the *separated* geometry ($\sigma_J = 3.27$, $\kappa = 8.89$): 0.11 versus 0.26 at 20% noise; both designs are full rank ($= 2$). **(b)** Experiment 4: σ_J (log) versus maximum eligible coherence across six wind providers $\times$ three sensor layouts (marker size $\propto$ coefficient error). Wind diversity lowers coherence (multi/real regulatory reach $0.13/0.16$) while a seeded random layout under steady wind collapses $\sigma_J \to 0$ and inflates coherence, producing the only nonzero errors (single/random: coherence 1.00, error 0.57). **(c)** Experiment 3: four predeclared backgrounds; the redundant basis is byte-identical to the primary (same span), while the labeled source-like *stress* basis drives absorption to 1 and σ_J and minimum visibility to 0: a too-flexible background can erase the attributed signal.

primary in every reported quantity, confirming that duplicate columns spanning the same subspace leave the projection, and therefore visibility, absorption, and recovery, unchanged. The labeled source-like *stress* basis is the cautionary case: it drives σ_J to 0, absorption to 1, and minimum visibility to 0, i.e. it absorbs the source-driven signal and destroys identifiability, even though the fitted residual is essentially unchanged. This is precisely how a too-flexible background can quietly erase the very signal being attributed. (Coefficient error is a misleading summary in the stress case: it *falls* to 0.78 from the primary's 2.61 precisely because the source-like basis has absorbed the signal, whereas the `none`, `primary`, and `redundant` bases give 2.36, 2.61, and 2.61.)

Wind Diversity and Sensor Geometry. Matched source, basis, and sensor sets are evaluated under constant, single-direction, diurnal, AR(1), multi-directional, and real gridded New Delhi wind. Sensor variants are the regulatory layout, seeded random layouts, and downwind-focused layouts. The same columns are required on both sides of every wind comparison. Separate historical and simulated window ensembles estimate the 5th, 50th, and 95th percentiles of σ_J, rank probabilities, coefficient weakness probabilities, pairwise ambiguity probabilities, and conservative component frequencies. Figure 1(b) crosses six wind providers with three sensor layouts (identical source–basis columns throughout; the design is rank-2 everywhere). Two effects dominate. First, wind diversity improves conditioning: multi-directional and real gridded New Delhi wind give the lowest maximum eligible coherence (0.13 and 0.16 at the regulatory layout), whereas a single steady direction pushes coherence toward 1. Second, geometry matters as much as wind: under a steady wind a seeded *random* layout collapses σ_J to ≈ 0 and inflates coherence, producing the only nonzero coefficient errors in the sweep (e.g. single-direction/random, $\sigma_J \approx 10^{-6}$, coherence 1.00, error 0.57), while the regulatory layout keeps σ_J well away from zero. Separate window ensembles confirm robustness across realized wind: historical (contiguous real-record slices) and simulated (AR(1)) both attain full numerical and effective rank with probability 1.0 and no pairwise ambiguity, differing only in their σ_J spread (5th/50th/95th percentiles $1.45/4.51/9.96$ historical versus $3.50/5.58/6.80$ simulated).

Missing-Source Adequacy. Controlled trials omit either a residual-visible source outside $\mathrm{span}[H_\Phi^{\mathrm{lag}}, Q]$ or an aligned source whose signature lies in that span. The refitted parametric-bootstrap test from Section 6.3 is evaluated for calibration and power. The aligned case demonstrates why non-rejection cannot certify inventory completeness. The adequacy study is run on the real New Delhi platform with a non-empty rank-4 primary background Q (fitted-design singular values $[120.6, 69.2, 45.8]$, so $\sigma_J = 45.8$). The refitted parametric bootstrap ($\alpha = 0.05$, 1000 replicates, 100 trials) is well-calibrated: with no omission the rejection rate is 0.05. It has full power against a residual-visible omitted source whose signature lies largely outside $\mathrm{span}[H_\Phi^{\mathrm{lag}}, Q]$ (out-of-span fraction 0.91; rejection rate 1.0). Crucially, an *aligned* omission (a genuinely omitted source whose signature is a nonnegative combination of the fitted columns and the background) is absorbed and not detected (rejection 0.05). The test is therefore one-sided: a rejection is evidence of model inadequacy (though not of its cause), whereas a non-rejection cannot certify adequacy, since an aligned omission is absorbed and passes undetected.

7.4 Observed New Delhi Results

The observed-data study has no source-activity ground truth and therefore reports geometry, residuals, uncertainty, and conservative groups rather than recovery error.

New Delhi Wind-Imputation Validation. The New Delhi wind field used throughout is produced by the kernel coordinate-query imputer described in Section 7.1. We also evaluated a learned wind model, but the sparse regulatory station network provides too little spatial support for it to learn useful spatial structure: it did not outperform a simple non-spatial city-mean baseline on held-out station–time vectors, and was not adopted. The kernel interpolator is used as the gridded field because, unlike the city mean, it is spatially resolved as the response operator requires. Dense real wind truth is unavailable, so gridded-field accuracy is instead assessed through the controlled synthetic-wind experiments (Experiment 4), where the true wind is known.

New Delhi Identifiability and Report Groups. The observed study is run as four consecutive one-week windows (2018-05-01 through 2018-05-28), each a self-contained fit reported on its own terms; results are not aggregated across windows. Figure 2(a) summarizes the identifiability geometry per week. Every week is full numerical rank (7 free coefficients after the fixed-zero admissibility mask), finite-conditioned, with low maximum eligible coherence (0.32–0.50), an empty weak set, no cross-source ambiguous pairs, and four singleton report components. Each source group is therefore individually identifiable in every week and no conservative merge is required.

New Delhi Proxy Apportionment and Uncertainty. Figure 2(b) reports the proxy apportionment for each week as a fraction of the fitted inventory-attributed sensor signal (denominator: the sum over groups of the L_1 fitted per-group sensor-signal magnitude); these are explicitly *not* physical-emission shares. Traffic is a single road-network group carrying diurnal slot temporal bases under the fixed-zero admissibility mask, so there is no per-slot `traffic_06` inventory to estimate; a group whose admissible temporal components are all fixed-zero in a window would be marked `unsupported` rather than assigned a substantive share (none were in these weeks). The apportionment swings substantially week to week: population-dominated in weeks 1–3 (shares 1.00, 0.81, 0.94), then brick-kiln-dominated in week 4 (0.93), which is precisely why the windows are reported separately rather than as one aggregate New Delhi estimate. Per-monitor fitted group contributions and their cells of origin (29/28 monitors per week) are not tabulated individually.

On observed data the fit uses only the 79–85% of rows with a valid $PM_{2.5}$ measurement (no imputation), subtracting each week's zero-source transported kriged initial-condition baseline (Pusa monitors averaged) under the gridded kernel wind field. Because no calibrated observation-noise model exists for the regulatory $PM_{2.5}$, the residual-adequacy test is reported as `uncalibrated` and no pass/fail verdict is claimed, per the contract of Section 6.3.

8 Discussion and Conclusion

We framed sparse-sensor source apportionment as an identifiable-resolution problem: known or proxy inventories restrict the source space and make attribution meaningful, but they do not guarantee that source groups are separable from sparse observations. Separability is governed by the projected lagged source–basis response matrix $\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}} \in \mathbb{R}^{N\times J}$ ($J = KB$): exact identifiability at the selected temporal-basis resolution requires $\operatorname{rank}(\widetilde{H}_\Phi) = J$, while robust attribution depends on its singular values, effective rank, coefficient visibility, background absorption, pairwise coherence, ray distance, and perturbation sensitivity. Because a model can fit sensor trajectories while producing non-unique coefficient estimates, prediction accuracy alone is not evidence of identifiability. IASA therefore reports a conservative grouping supported by the sensing system, estimating source contributions where the data support them and coarsening or qualifying attribution (deterministic connected report groups, weak-coefficient flags, retained trigger pairs) where the projected response does not. The same view guides design: sensors, wind diversity, and background choices should be judged by whether they improve the visibility, conditioning, and coherence of coefficient fingerprints, not merely spatial coverage.

Several limitations bound these claims. The theory is linear in source–basis coefficients conditional on fixed wind, transport, temporal basis, inventory, and background; the selected temporal basis limits recoverable activity patterns (modeling traffic as one spatial map with diurnal components assumes a time-invariant congestion pattern); missing source categories may be absorbed into available inventories or background, a misspecification risk rather than identifiable attribution; a calibrated residual bootstrap can reject an inadequate model but cannot prove inventory completeness, especially when an omitted signature lies in the span of the fitted response and background; transport ensembles quantify uncertainty under their sampling model whereas alternative inventories are separate robustness scenarios that must not be pooled into intervals; and the gridded wind field is only as reliable as the sparse station network and its held-out validation. Extending the open-boundary puff response with richer meteorology, vertical mixing, deposition, chemistry, and boundary inflow, augmenting the per-sensor footprints of Section 5.4 with explicit spatial diagnostics, and jointly estimating temporal basis, transport, and inventory under identifiability control are natural next steps. By separating fitting the sensor signal from determining which source claims the sensing system supports, the identifiability view provides a concrete basis for more reliable policy-facing apportionment.

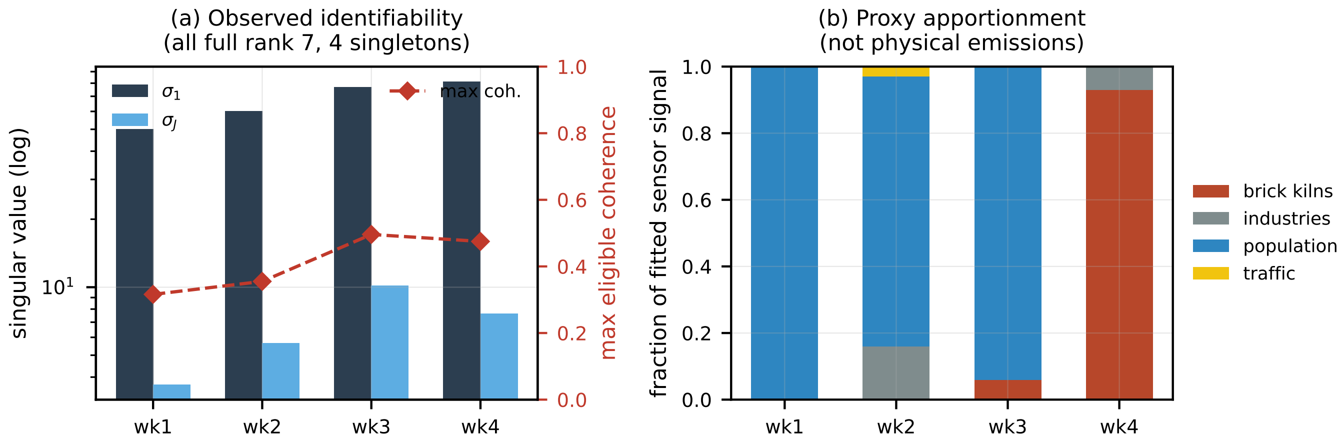

Figure 2: Observed New Delhi study, four consecutive one-week windows (2018-05-01 through 2018-05-28), each fitted and reported independently. **(a)** Identifiability geometry: every week is full numerical rank (7 free coefficients after the fixed-zero admissibility mask, four singleton report components), with σ_1 rising from 50.1 to 81.2, σ_J from 3.71 to 10.15 (week 3), and low maximum eligible coherence (0.32–0.50). **(b)** Proxy apportionment as a fraction of the fitted inventory-attributed sensor signal (*not* physical-emission shares): the mixture swings from population-dominated in weeks 1–3 to brick-kiln-dominated in week 4, which is why the windows are reported separately rather than as one aggregate estimate.

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Table 1: Indices and objects in the temporal-basis apportionment model.

Symbol	Shape or range	Meaning
k	$1, \ldots, K$	source-group index
b	$1, \ldots, B$	temporal-basis index
$j = (k, b)$	$1, \ldots, J$	source-major, basis-minor coefficient index
Φ	$T \times B$	nonnegative source-activity basis
C	$K \times B$	nonnegative source–basis coefficients
$\mathbf{c}$	$J = KB$	source-major vectorization of C
H_{Φ}^{lag}	$N \times J$	observed-row coefficient fingerprints
$\widetilde{H}_{\Phi}$	$N \times J$	background-projected fingerprints
$\widetilde{\mathbf{h}}_{kb}$	N	fingerprint for coefficient c_{kb}

A Temporal-Basis Notation and Diagnostic Conventions

The constant-activity model uses $B = 1$, $\phi_1(t) = 1$, and $c_{k1} = \theta_k$. In the general model, diagnostics are computed for the J coefficient fingerprints. Source-level activity trajectories are reconstructed as

$$\widehat{\theta}_k(t) = \sum_{b=1}^{B} \widehat{c}_{kb}\phi_b(t);$$

fingerprint columns are not summed to create an artificial source-level diagnostic vector.

For a source-level ambiguity summary, IASA reports the largest eligible coherence between coefficients belonging to different sources and retains the source–basis pair that attains it. Coherence and ray distance are undefined for pairs involving a coefficient with $v_j \le \tau_v$; tables display these entries as N/A, exclude them from pairwise extrema, and report the weak coefficient separately.

B Detailed Identifiability Proofs

Proof of Proposition 5.2. Let $A = \widetilde{H}_{\Phi} \in \mathbb{R}^{N \times J}$. If $\operatorname{rank}(A) = J$, then the rank-nullity theorem gives $\dim \ker(A) = J - \operatorname{rank}(A) = 0$. Hence $\ker(A) = \{\mathbf{0}\}$. If two feasible coefficient vectors $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}_+^J$ produce the same projected observation,

$$A\mathbf{c}_1 = A\mathbf{c}_2,$$

then $A(\mathbf{c}_1 - \mathbf{c}_2) = 0$. Since the only vector in the kernel is $\mathbf{0}$, we have $\mathbf{c}_1 - \mathbf{c}_2 = \mathbf{0}$, and therefore $\mathbf{c}_1 = \mathbf{c}_2$. Thus the coefficient vector is identifiable over the nonnegative feasible set.

Conversely, suppose $\operatorname{rank}(A) < J$. By rank-nullity, $\dim \ker(A) > 0$, so there exists a nonzero vector $\Delta\mathbf{c} \in \ker(A)$. Choose any $\beta > \|\Delta\mathbf{c}\|_\infty$ and set $\mathbf{c} = \beta\mathbf{1}$. Then $\mathbf{c} \in \mathbb{R}_+^J$, and $\mathbf{c} + \Delta\mathbf{c} \in \mathbb{R}_+^J$ because every component satisfies

$$\beta + \Delta c_j \ge \beta - \|\Delta\mathbf{c}\|_\infty > 0.$$

Table 2: Identifiability diagnostics and reporting outputs produced with source activity estimates. The lower block lists the set- and graph-valued outputs that accompany the scalar diagnostics.

Symbol	Name	Interpretation	Action
$r_{\mathrm{num}}(\widetilde{H}_\Phi)$	numerical rank	exact separability	require rank J
$\sigma_J(\widetilde{H}_\Phi)$	padded min. singular value	noise robustness	flag zero/small
$\kappa(\widetilde{H}_\Phi)$	condition number	error amplification	∞ if deficient
r_{eff}	effective rank	noise-level resolution	reduce grouping
v_j	coefficient visibility	measurable effect	flag weak
a_j	background absorption	signal lost to Q	warn after projection
ρ_{ij}	pairwise coherence	pairwise ambiguity	merge coherent
d_{ij}^{ray}	ray distance	ray separation	N/A if weak
$\mathcal{W}$	weak set	undetectable coefficients	flag, no edges
$\mathcal{A}$	ambiguous-pair set	confusable pairs	source edges
$\mathcal{E}_{\mathrm{src}}$	source ambiguity graph	source-level ambiguity	retain triggers
$\mathcal{G}$	connected report groups	identifiable resolution	report grouped
n/a	global unresolved warning	deficient, no edge	warn globally

The two vectors are distinct because $\Delta\mathbf{c} \neq 0$, but they produce the same projected observation:

$$A(\mathbf{c} + \Delta\mathbf{c}) = A\mathbf{c} + A\Delta\mathbf{c} = A\mathbf{c}.$$

Therefore uniform identifiability over $\mathbb{R}_+^J$ fails whenever $\operatorname{rank}(A) < J$. This proves the equivalence.

Proof of Proposition 5.3. Again let $A = \widetilde{H}_\Phi$, and assume A has full column rank J. The projected observation model is

$$\widetilde{Y} = A\mathbf{c} + \widetilde{E}.$$

For unconstrained least squares,

$$\widehat{\mathbf{c}}_{\mathrm{LS}} = \arg\min_{\mathbf{z}} \|\widetilde{Y} - A\mathbf{z}\|_2^2 = A^\dagger \widetilde{Y},$$

where $A^\dagger$ is the Moore–Penrose pseudoinverse. Substituting the data model gives

$$\widehat{\mathbf{c}}_{\mathrm{LS}} = A^\dagger(A\mathbf{c} + \widetilde{E}) = \mathbf{c} + A^\dagger \widetilde{E},$$

because $A^\dagger A = I_J$ for a full-column-rank matrix. Hence

$$\|\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c}\|_2 = \|A^\dagger \widetilde{E}\|_2 \leq \|A^\dagger\|_2 \|\widetilde{E}\|_2.$$

The singular values of $A^\dagger$ are the reciprocals of the nonzero singular values of A, so $\|A^\dagger\|_2 = 1/\sigma_J(A)$. Therefore

$$\|\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c}\|_2 \leq \frac{\|\widetilde{E}\|_2}{\sigma_J(A)}.$$

For nonnegative least squares, let $\widehat{\mathbf{c}}$ minimize $\|\widetilde{Y} - A\mathbf{z}\|_2^2$ over $\mathbf{z} \geq 0$. Since the true $\mathbf{c} \geq 0$, it is feasible. Optimality of $\widehat{\mathbf{c}}$ therefore implies

$$\|\widetilde{Y} - A\widehat{\mathbf{c}}\|_2 \leq \|\widetilde{Y} - A\mathbf{c}\|_2 = \|\widetilde{E}\|_2.$$

Let $\Delta = \widehat{\mathbf{c}} - \mathbf{c}$. Using $\widetilde{Y} = A\mathbf{c} + \widetilde{E}$,

$$\widetilde{Y} - A\widehat{\mathbf{c}} = \widetilde{E} - A\Delta.$$

Thus

$$\begin{aligned} \|A\Delta\|_2 &= \|\widetilde{E} - (\widetilde{E} - A\Delta)\|_2 \\ &\leq \|\widetilde{E}\|_2 + \|\widetilde{E} - A\Delta\|_2 \leq 2\|\widetilde{E}\|_2. \end{aligned}$$

Full column rank gives the lower singular-value inequality

$$\|A\Delta\|_2 \geq \sigma_J(A)\|\Delta\|_2.$$

Combining the two inequalities yields

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 = \|\Delta\|_2 \leq \frac{2\|\widetilde{E}\|_2}{\sigma_J(A)}.$$

This proves the stated NNLS robustness bound.

Ray distance for nonweak fingerprints (Equation 11). Fix nonweak projected fingerprints $\widetilde{\mathbf{h}}_i, \widetilde{\mathbf{h}}_j$, so $\widetilde{\mathbf{h}}_j \neq 0$. The ray distance minimizes

$$g(\alpha) = \frac{\|\widetilde{\mathbf{h}}_i - \alpha\widetilde{\mathbf{h}}_j\|_2^2}{\|\widetilde{\mathbf{h}}_i\|_2^2}$$

over $\alpha \in \mathbb{R}$. This is strictly convex in α, and $g'(\alpha) = 0$ gives $-2\widetilde{\mathbf{h}}_j^\top(\widetilde{\mathbf{h}}_i - \alpha\widetilde{\mathbf{h}}_j) = 0$, i.e.

$$\alpha^\star = \frac{\widetilde{\mathbf{h}}_i^\top \widetilde{\mathbf{h}}_j}{\|\widetilde{\mathbf{h}}_j\|_2^2}.$$

Substituting,

$$\|\widetilde{\mathbf{h}}_i - \alpha^\star\widetilde{\mathbf{h}}_j\|_2^2 = \|\widetilde{\mathbf{h}}_i\|_2^2 - \frac{(\widetilde{\mathbf{h}}_i^\top \widetilde{\mathbf{h}}_j)^2}{\|\widetilde{\mathbf{h}}_j\|_2^2},$$

and dividing by $\|\widetilde{\mathbf{h}}_i\|_2^2$ yields $(d_{ij}^{\mathrm{ray}})^2 = 1 - \rho_{ij}^2$, so $d_{ij}^{\mathrm{ray}} = \sqrt{1 - \rho_{ij}^2}$. Minimizing over signed α makes the sign inside ρ_{ij} immaterial.

Proof of Proposition J.1. Let $\widehat{A} = \widehat{\widetilde{H}}_\Phi$ have full column rank and let $A^\star = \widetilde{H}_\Phi^\star$ with $\|\widehat{A} - A^\star\|_2 \leq \delta_H$. The data obey $\widetilde{Y} = A^\star\mathbf{c} + \widetilde{E} = \widehat{A}\mathbf{c} + [\widetilde{E} + (A^\star - \widehat{A})\mathbf{c}]$. The least-squares estimate using $\widehat{A}$ is $\widehat{\mathbf{c}} = \widehat{A}^\dagger\widetilde{Y}$, and $\widehat{A}^\dagger\widehat{A} = I_J$, so

$$\widehat{\mathbf{c}} - \mathbf{c} = \widehat{A}^\dagger\big[\widetilde{E} + (A^\star - \widehat{A})\mathbf{c}\big].$$

Taking norms with $\|\widehat{A}^\dagger\|_2 = 1/\sigma_J(\widehat{A})$ and $\|(A^\star - \widehat{A})\mathbf{c}\|_2 \leq \delta_H\|\mathbf{c}\|_2$ gives

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 \leq \frac{\|\widetilde{E}\|_2 + \delta_H\|\mathbf{c}\|_2}{\sigma_J(\widehat{A})},$$

which is Equation 21.

C Implementation Invariants

The response is constructed first on the complete time-major, sensor-minor row index. The $PM_{2.5}$ observation mask then selects the same ordered rows from Y, H_Φ^{lag}, Q, and metadata. Source–basis columns are always source-major and basis-minor. Any row or column metadata mismatch is a hard error.

The response construction uses integer source-cell centers, half-cell domain extents, fractional final advection substeps, a positive minimum dispersion age, direct sensor evaluation, finite lag, and complete removal after center exit. Kernel boundary/truncation loss and whole-release exit loss are recorded separately. Retained and dropped kernel mass sum to emitted kernel mass within the recorded quadrature tolerance; sensor observations are never summed as a mass proxy. The default puff baseline has zero source and zero initial state.

Normal backgrounds have effective rank at most eight and do not use inventory or response columns. Thin-SVD projection stores the numerical tolerance, singular values, retained U_r, dependent input columns, removed components, and orthogonality/idempotence residuals. A source-like column is permitted only when the run is labeled as a stress test.

The primary Q, lag-candidate grid, and fixed-zero coefficient set are declared before fitting. A fixed-zero mask removes columns before diagnostics and fitting and stores both reduced-to-original and original-to-reduced index maps. Fitted near-zero coefficients never modify the mask. Adjacent lag responses use identical rows and columns; the default convergence threshold is $\tau_L = 10^{-3}$.

D Sanity Gates

Before paper experiments, the implementation must pass six small deterministic gates built from public APIs:

1. **S1: response.** On a 16×16 grid with four compact sources, four sensors, impulse/constant bases, and seed 123, verify response shape, downwind arrival, open-boundary exit, no wraparound, dispersion anisotropy, mass accounting, metadata, and exact repeatability.
2. **S2: projection.** Use the S1 response with a rank-four normal background to verify exact ordering, background removal, orthogonality, idempotence, redundant-column invariance, and the labeled source-like stress behavior.
3. **S3: diagnostics.** Verify orthogonal, duplicate, and weak matrices, including padded zeros, σ_J, infinite deficient condition status, weak-pair N/A values, and a matched multi-source eastward versus two-direction wind comparison.
4. **S4: fit.** Recover known nonnegative coefficients in noiseless/noisy well-conditioned cases, verify projected-FISTA KKT convergence, preserve exact nonnegativity, and recover the duplicate-pair sum when the individual split is nonunique.
5. **S5: report groups.** Verify the duplicate-source connected component, an A–B–C chain with both trigger edges retained, separated-source non-edge, independent weak flag, conservative-group flag, and summed group contribution.
6. **S6: end to end.** Build sources, wind, response, background, masked observations, diagnostics, fit, uncertainty, and report groups with one command and write a compact provenance-complete summary.

Task 10 experiments begin only after S1–S6 and the PyTorch CPU/CUDA parity checks pass. Two further gates support the results. **S7 (adequacy calibration)** runs a statistically calibrated study of the residual-adequacy test on a controlled orthonormal design, checking that the null rejection rate matches α, that the null p-values are uniform, and that power increases monotonically with omission strength. A final **reporting gate** exercises the evaluation layer that emits the result tables in Section 7, verifying our reporting rules: undefined weak-pair metrics stay null, grouped metrics accompany any non-singleton report component, an A–B–C merge chain retains both trigger edges, percentages are labeled fractions of fitted sensor signal, and an uncalibrated run never reports an adequacy pass.

E PyTorch Solver and Uncertainty Protocol

After CSV/NPZ ingestion, numerical computation uses PyTorch tensors on an explicit device and dtype. The nonnegative coefficient objective is

$$f(\mathbf{c}) = \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda \|\mathbf{c} - \mathbf{c}_0\|_2^2, \qquad \mathbf{c} \geq 0. \tag{12}$$

Projected FISTA uses a Lipschitz step derived from σ_1 or a recorded power-iteration estimate, clamps every iterate to the nonnegative orthant, and restarts acceleration whenever the objective increases. Termination requires the configured projected-gradient/KKT residual and relative objective-change tolerances, or reports that the iteration cap was reached. The fit artifact stores convergence status, iteration count, final KKT residual, objective summary, device, and dtype.

Active-set covariance uses PyTorch linear algebra on $\widetilde{H}_{\Phi,\mathcal{A}}^\top \widetilde{H}_{\Phi,\mathcal{A}} + \lambda I$. Wind-ensemble and bootstrap refits are batched on the same device when shapes permit. Undefined weak-pair metrics are serialized as JSON `null`, never NaN or a false numerical distance.

Uncertainty artifacts distinguish `ensemble_kind=transport` from `ensemble_kind=inventory`. Only the former may be summarized as an ensemble uncertainty interval; inventory members remain named robustness scenarios, and an aggregation request that mixes the two kinds is an error.

F Adequacy and Ensemble Result Contracts

A residual-adequacy result stores `calibration_status`, T_{res}, bootstrap quantile, Monte Carlo p-value, α, replicate count, and noise-model provenance. Paper runs use $\alpha = 0.05$ and 1000 refitted replicates. If the externally calibrated noise covariance is absent or invalid, numerical test fields are null and the status is `uncalibrated`; no boolean pass is emitted. Sensor, time, and autocorrelation residual summaries are retained in either case.

A lag-selection result stores every candidate L, adjacent convergence ratio, selected L, τ_L, physical grid rationale, row count, and diagnostic/group summaries. A wind-distribution result stores the sampled-window provenance; 5th, 50th, and 95th percentiles of σ_J; full numerical and effective-rank probabilities; coefficient weakness probabilities; pairwise ambiguity probabilities; and conservative-component frequencies.

G Reproducibility and Artifact Provenance

Every paper run records the container image, repository revision, PyTorch and CUDA versions, device model, dtype, deterministic-algorithm settings, random seeds, source files and crop, per-source scales, wind-imputer architecture, held-out wind split, query grid, gridded wind product, and ensemble provenance, inventory identifier/hash/version, observation mask, response and transport configuration, wind realization, dispersion settings, background columns and rank tolerance, lag protocol, fixed-zero mask and index mapping, diagnostic thresholds, ensemble kind, solver settings, noise-model provenance, and parent artifact hashes. Generated wind products, response matrices, checkpoints, and experiment outputs are written to generated-data or log paths and are not treated as immutable source files.

Controlled-run artifacts contain true and estimated coefficients, reconstructed activities, response/projection results, diagnostics, uncertainty, report groups, and recovery metrics. Observed New Delhi artifacts omit unavailable ground truth and instead contain observed-row counts, fitted trajectories, raw and projected residuals, normalized proxy activities/contributions, uncertainty, weak/ambiguous flags, trigger pairs, and group summaries. A percentage is emitted only with an explicit denominator identifying it as a fraction of fitted inventory-attributed sensor signal.

H Extended Discussion on Related Work

This appendix expands the four strands summarized in Section 2.

Air-pollution source apportionment. Source apportionment has a long history in air-quality science, with receptor-oriented methods estimating source contributions from ambient chemical or physical measurements and source-oriented methods propagating emissions inventories through atmospheric transport models. Classical receptor frameworks include chemical mass balance and factor-analytic approaches such as positive matrix factorization (Watson et al. 2002; Paatero and Tapper 1994; Hopke 2016). Reviews and harmonization efforts have clarified best practices for receptor models, source-oriented models, and their combined use, while also documenting practical sensitivity to source profiles, factor interpretation, chemical resolution, and model comparability (Viana et al. 2008; Belis et al. 2013, 2019; Mircea et al. 2020; Hopke et al. 2020). These sensitivities arise because ambient measurements provide only indirect mixtures of source contributions: source profiles may be incomplete or locally unavailable, tracers and proxy inventories can be collinear, factor decompositions may admit multiple plausible interpretations, and transport or inventory assumptions can map distinct source configurations to similar observations. This ambiguity is especially visible in settings such as India, where studies can produce divergent source-contribution estimates for the same city because of limited local profiles, collinear tracers, and methodological differences (Pant and Harrison 2012); global reviews further show that policy-relevant source categories such as traffic, industry, dust, and domestic combustion are routinely reported from heterogeneous apportionment studies (Karagulian et al. 2015). Our work is complementary to this literature: rather than proposing another receptor model or emissions inventory, we analyze when known or proxy source maps are distinguishable from sparse concentration sensors after wind-conditioned transport, finite lags, background correction, and noise.

Inverse modeling and physics-constrained learning. Estimating latent states or parameters from partial observations is central to inverse problems and data assimilation (Tarantola 2005; Engl, Hanke, and Neubauer 1996). Classical approaches such as Kalman filtering, ensemble Kalman filters, and variational data assimilation use dynamical models together with priors, covariance assumptions, and observation models to infer hidden quantities from measurements (Wang et al. 2023; Carrassi et al. 2018). More recent physics-informed and operator-learning methods, including PINNs and neural operators, incorporate governing equations or physics-motivated structure into flexible learning systems for parameter estimation and system identification (Raissi, Perdikaris, and Karniadakis 2019; Li et al. 2024, 2020; Bhardwaj, Balashankar, and Subramanian 2025). These methods motivate our use of a transport-structured, PyTorch-based constrained fitting backend, but they do not by themselves resolve the identifiability question. In sparse sensing regimes, the map from latent source activities, transport parameters, and background components to sensor observations can be many-to-one: different parameterizations may fit the same measurements after optimization. Our focus is therefore complementary to calibration and physics-constrained learning methods. Rather than asking only how to fit the projected sensor data, we characterize when the fitted source–basis coefficients are identifiable from the projected lagged response matrix and when the result should be weakened, qualified, or grouped.

Sensor-space spatio-temporal learning. Graph neural networks, transformers, and sequence models have been widely used to model spatio-temporal processes from sensor data, including traffic, environmental, and air-quality time series (Li et al. 2017; Yu, Yin, and Zhu 2017; Wu et al. 2019; Chen et al. 2023; Iyer et al. 2022; Bhardwaj et al. 2025; Bhardwaj, Balashankar, and Subramanian 2025). These methods are relevant to our setting because they operate in the same sparse-sensor observational regime. However, their usual objective is forecasting, interpolation, or representation learning over sensor measurements rather than attribution to physically meaningful source inventories. A sensor-space model can predict observed concentrations well while leaving unresolved whether distinct source groups would produce distinguishable wind-conditioned fingerprints at the deployed sensors. Our work is therefore complementary: instead of learning an unconstrained predictive representation of the sensor field, we analyze the identifiability of declared source–basis contributions under a specified transport, background, lag, and noise model.

Observability and system identification. The question of whether latent quantities can be recovered from observations is closely related to classical notions of observability in control theory (Chen and Chen 1984). For linear dynamical systems, observability characterizes whether the internal state can be uniquely determined from output trajectories. Extensions to nonlinear systems and PDEs have been studied in system identification and inverse-problem literature (Ljung et al. 1987; Banks and Kunisch 2012; Colton et al. 1990), often under assumptions of sufficient measurement richness. We bring this perspective to inventory-based air-pollution source apportionment under sparse spatial sensing. Rather than asking whether a full latent state or physical model is observable, we ask whether declared source–basis contributions have distinct projected sensor-time fingerprints after wind-conditioned transport, finite lags, background correction, and noise. This leads to diagnostics tailored to the apportionment task, including rank and singular-value stability, source visibility, pairwise coherence, background absorption, and conservative grouping, together with a real-world instantiation that reports the attribution resolution supported by the available sensors, inventories, and transport assumptions.

I Transport Response Construction and Fit Details

This appendix gives the construction, background, and estimation details deferred from Section 4.

Wind imputer query and unit conversion. The observed station vectors and masks are supplied to a coordinate-query wind imputer: given sparse station observations $\{(\mathbf{z}_i, t, \mathbf{u}_{i,t}, m_{i,t})\}$, it predicts transport vectors at arbitrary spatial query locations and times. For the IASA response operator, we query the imputer on every response-grid cell $\mathbf{x}_g$ and hour t, producing

$$\widehat{\mathbf{w}}_t(\mathbf{x}_g) = \widehat{f}_\omega(\mathbf{x}_g, t; \{(\mathbf{z}_i, t', \mathbf{u}_{i,t'}, m_{i,t'})\}) \,,\ g = 1, \ldots, n. \tag{13}$$

This yields a gridded wind field $\widehat{W} \in \mathbb{R}^{T\times n\times 2}$, rather than a station-only or city-averaged sequence. Real-data validation is performed by masking observed station vectors and measuring held-out station-time error; dense city-wide wind truth is not assumed. Meteorological speed must be converted to grid displacement before puff advection. For response timestep Δt_s seconds and cell dimensions $\Delta x_\mathrm{m}, \Delta y_\mathrm{m}$, we use

$$v_x^{\mathrm{grid}}(\mathbf{x}_g, t) = \frac{\widehat{U}_x(\mathbf{x}_g, t)\Delta t_\mathrm{s}}{\Delta x_\mathrm{m}}, \qquad v_y^{\mathrm{grid}}(\mathbf{x}_g, t) = \frac{\widehat{V}_y(\mathbf{x}_g, t)\Delta t_\mathrm{s}}{\Delta y_\mathrm{m}}. \tag{14}$$

Every response artifact records these scales, the coordinate convention, the wind units, and the exact converted sequence.

Wind-field ensembles. The imputed gridded wind field is an estimated transport input, not ground truth. To propagate wind-imputation uncertainty, we construct wind-field ensembles $\{\widehat{\mathbf{w}}_t^{(r)}(\mathbf{x}_g)\}_{r=1}^R$ using held-out-calibrated prediction error, station bootstrap resampling, checkpoint ensembles, or perturbations matched to validation residuals. Each ensemble member is queried on the same response grid and converted to grid displacement before response construction. The resulting response matrices are tagged as transport ensembles and are kept separate from inventory-robustness scenarios.

Puff dynamics and dispersion. We instantiate the response with a Gaussian puff approximation. For a unit release from source cell i at location $\mathbf{r}_i$ and release time τ, the puff center is initialized as $\mathbf{z}_i(\tau) = \mathbf{r}_i$ and advected by the interpolated wind field, $d\mathbf{z}_i(a)/da = \widehat{\mathbf{w}}_a(\mathbf{z}_i(a))$, $a \in [\tau, t]$. Source cells have integer centers and domain extents $[-0.5, N_x - 0.5] \times [-0.5, N_y - 0.5]$. With substep size δa, the trajectory is discretized as $\mathbf{z}_i^{r+1} = \mathbf{z}_i^r + \delta a\, \widehat{\mathbf{w}}_{a_r}(\mathbf{z}_i^r)$. The final substep is shortened so that the trajectory lands exactly at the requested observation time. If the center exits Ω, the whole remaining puff is removed and never reflected, wrapped, clamped, or reinserted. Otherwise its contribution is spread with an anisotropic Gaussian kernel

$$K_\psi(\mathbf{x}; \mathbf{z}_i, \Sigma_i) = \frac{\exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{z}_i\|^2_{\Sigma_i^{-1}}\right)}{2\pi|\Sigma_i|^{1/2}}, \tag{15}$$

where ψ denotes dispersion parameters. A simple covariance model is

$$\Sigma_i(t,\tau) = R_i(t,\tau)\begin{bmatrix}\sigma_\parallel^2(t-\tau) & 0\\ 0 & \sigma_\perp^2(t-\tau)\end{bmatrix} R_i(t,\tau)^\top, \tag{16}$$

with R_i aligned to the mean wind direction along the trajectory. When the mean wind norm is below the numerical threshold, the downwind direction is undefined; the implementation uses the positive-x axis as a deterministic tie-breaker for the anisotropic covariance orientation, recorded in the response metadata and affecting only near-calm releases. The age in this covariance is lower-bounded by a positive minimum dispersion time, so same-time releases remain finite: effective age $= \max(t - \tau, t_{\min})$.

Sensor evaluation and loss bookkeeping. Response-matrix entries are computed by evaluating the kernel directly at sensor coordinates:

$$\left[OG^\partial_{t,\tau}(\widehat{\mathbf{w}}_{\tau:t}; \psi)\mathbf{e}_i\right]_s = \chi_i(t,\tau) K_\psi(\mathbf{x}_s; \mathbf{z}_i(t), \Sigma_i(t,\tau)). \tag{17}$$

Here $\chi_i(t,\tau)$ is an open-boundary survival indicator for a release from source cell i at release time τ: it is one while the advected kernel center remains inside the modeled domain and zero after the release has exited. Boundary and truncation losses are recorded only as diagnostic quantities and are never used to renormalize the sensor response. Boundary loss denotes the portion of the Gaussian kernel lying outside the modeled spatial domain, while truncation loss denotes the portion omitted by evaluating retained mass only over a finite kernel support; both are estimated separately by grid quadrature over the response grid. These kernel losses are distinct from whole-release exit loss, where the advected release center leaves the open domain and no later sensor contribution is assigned to that release. Because the response matrix evaluates concentration at sensor points rather than integrating over area-weighted grid cells, summing entries across sensors or repeated sensor-time observations does not estimate retained pollutant mass.

Baseline policy and differentiability. Response matrices are constructed for source-induced increments relative to a declared initial-condition baseline. The baseline policy is fixed before source fitting, recorded with the response artifact, and applied consistently to the observations and response rows. Application-specific choices, such as whether to subtract a zero-source transported initial field or to use a zero-initial-state response, are part of the experimental instantiation rather than the generic response definition. The open-boundary lagged plume response is implemented in PyTorch; tensor operations retained in the graph can be differentiated, but discrete exit events and reporting logic are not assigned gradients.

Background basis and implicit projection. The normal background basis may contain a global constant, centered and scaled elapsed-time polynomials, local-clock daily sine/cosine harmonics, reference-coded day effects, standardized regional sensor-coordinate trends, reference-coded sensor offsets, and a row-aligned user basis. It depends only on timestamps, day labels, sensor identity, and sensor coordinates, and its effective rank is capped at eight. Once the primary Q is fixed, its coefficients are estimated implicitly by projection, or explicitly in the joint fit; each alternative source-independent basis reports the resulting visibility, absorption, rank, conditioning, and ambiguity components. Letting $Q = U\Sigma V^\top$ be a thin singular value decomposition and retaining columns U_r above the recorded rank tolerance, projection is applied implicitly,

$$P_Q^\perp X = X - U_r(U_r^\top X), \tag{18}$$

without constructing an $N \times N$ matrix. The row metadata for Y, H_Φ^{lag}, and Q must be identical: each row must refer to the same timestamp, sensor identity, and sensor index in the same order before projection is applied.

Regularizers and joint fit. The regularizer in Equation 7 can be ridge stabilization $R(\mathbf{c}) = \|\mathbf{c}\|_2^2$ or a weak prior around inventory-derived activity levels $R(\mathbf{c}) = \|\mathbf{c} - \mathbf{c}_0\|_2^2$. Equivalently, one may fit background and source coefficients jointly,

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\gamma}}) = \arg\min_{\mathbf{c}\geq 0, \boldsymbol{\gamma}} \|Y - H_\Phi^{\mathrm{lag}}\mathbf{c} - Q\boldsymbol{\gamma}\|_2^2 + \lambda R(\mathbf{c}). \tag{19}$$

The projected formulation is used for identifiability because it exposes the part of each source–basis fingerprint that remains after background correction.

Constrained end-to-end refinement. IASA first constructs a fixed response matrix from the pretrained wind field and default dispersion parameters, fits source activities, and computes identifiability diagnostics for that declared response. It then optionally performs a constrained local refinement of the wind field, dispersion parameters, source coefficients, and background coefficients, initialized at the fixed-response solution and allowed only to make small physically constrained corrections, so that improved sensor fit does not come from arbitrary changes to transport. Let ϕ denote the wind-imputer parameters and ψ the dispersion parameters of the puff response (such as $\sigma_\parallel$, $\sigma_\perp$, and the minimum dispersion age), with ϕ_0 and ψ_0 their pretrained or default values. We use R_{sm} for a predeclared wind-field smoothness penalty (temporal or spatial squared differences) and Ψ_{phys} for the physically admissible dispersion-parameter set. The refinement objective is

$$\begin{aligned} \mathcal{L}_{\mathrm{refine}} = & \|Y - H_\Phi^{\mathrm{lag}}(\phi,\psi)\mathbf{c} - Q\boldsymbol{\gamma}\|_2^2 + \lambda_\theta R(\mathbf{c}) \\ & + \lambda_w \|\widehat{\mathbf{w}}_{1:T}^{\phi} - \widehat{\mathbf{w}}_{1:T}^{\phi_0}\|_2^2 + \lambda_\psi \|\psi - \psi_0\|_2^2 \\ & + \lambda_{\mathrm{sm}} R_{\mathrm{sm}}(\widehat{\mathbf{w}}^{\phi}), \end{aligned} \tag{20}$$

subject to $\mathbf{c} \geq 0$, $\psi \in \Psi_{\mathrm{phys}}$, and $\|\widehat{\mathbf{w}}_{1:T}^{\phi} - \widehat{\mathbf{w}}_{1:T}^{\phi_0}\|_\infty \leq \epsilon_w$. The refined response is accepted only as a constrained local correction (acceptance criteria in Appendix L) and is reported with its own response diagnostics. Without these restrictions, wind and plume parameters could fit sparse pollution sensors while weakening the physical and source-level interpretation of $\widehat{\mathbf{c}}$ and the reconstructed activities.

J Response-Matrix Perturbations

This appendix expands Section 5.3. Write the projected response discrepancy as $\Delta_H = \Delta_{\mathrm{tr}} + \Delta_{\mathrm{inv}} + \Delta_{\mathrm{int}}$, where Δ_{tr} is the transport term, Δ_{inv} the inventory term, and Δ_{int} their interaction. The decomposition is used to keep uncertainty honest: Δ_{tr} is propagated through transport ensembles and may become an uncertainty interval, whereas Δ_{inv} is reported only as named robustness scenarios and is never pooled into an interval, and Δ_{int} is recorded rather than pooled separately. Let $\widetilde{H}_\Phi^\star$ be the true projected response and $\widehat{\widetilde{H}}_\Phi$ the constructed one, with $\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \leq \delta_H$. Then $\widetilde{Y} = \widehat{\widetilde{H}}_\Phi\mathbf{c} + [\widetilde{E} + (\widetilde{H}_\Phi^\star - \widehat{\widetilde{H}}_\Phi)\mathbf{c}]$, so response error acts like additional observation error.

Proposition J.1 (Perturbation bound)**.** *Assume $\widehat{\widetilde{H}}_\Phi$ has full column rank and $\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \leq \delta_H$. The unconstrained least-squares estimate using $\widehat{\widetilde{H}}_\Phi$ satisfies*

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 \leq \frac{\|\widetilde{E}\|_2 + \delta_H\|\mathbf{c}\|_2}{\sigma_J(\widehat{\widetilde{H}}_\Phi)}. \tag{21}$$

A short proof, reusing the pseudoinverse bound of Proposition 5.3, appears in Appendix B. The same norm bound applies to all three discrepancy terms, but their interpretations are not interchangeable. Wind and dispersion ensembles quantify transport uncertainty and may support uncertainty intervals under the ensemble model. Alternative inventory maps, locations, labels, or scales are inventory robustness scenarios, not confidence intervals. They are reported separately, and source-specific attribution remains conditional on the supplied inventory. Either error type is most damaging when the projected response is already ill-conditioned.

K Per-Sensor Source Footprints

This appendix expands Section 5.4. For an observed row (s,t), the background-corrected fit decomposes exactly as $\widetilde{y}_{s,t} = \sum_{k,b} \widetilde{H}_{\Phi,(s,t),(k,b)}\,\widehat{c}_{kb}$, so the contribution of source k to sensor s over the record is $\widehat{Y}_k^{(s)} = \sum_t \sum_b H_{\Phi,(s,t),(k,b)}^{\mathrm{lag}}\widehat{c}_{kb}$.

Table 3: IASA thresholds, their meaning, and how they are set. None is optimized on source-recovery error.

Threshold	Meaning	How set
τ_{num}	numerical rank tol.	$\max(N, J)\epsilon_{\text{mach}}\sigma_1$
τ_σ	effective-rank floor	noise scale, e.g. $\sigma_e\sqrt{N}$
τ_v	visibility floor	$\text{SNR}_{\min}\,\sigma_e$
τ_ρ	ambiguity coherence	predeclared, near 1
τ_ρ^{ref}	refinement coherence	default $\tau_\rho^{\text{ref}} = \tau_\rho$
ϵ_w	wind-drift cap	physical wind tolerance

The projected form isolates the identifiable part; the unprojected form gives the raw fitted sensor contribution before background removal. The spatial origin of a sensor's signal is the pullback of its response row onto the source grid: for sensor s at time t, $F_{s,t}(i) = \sum_{\ell\in\mathcal{L}_t}[OG^{\partial}_{t,t-\ell}(\widehat{\mathbf{w}}_{t-\ell:t})]_{s,i} \geq 0$ and $F_s(i) = \sum_t F_{s,t}(i)$, a nonnegative field over source cells i recording which upwind cells influence sensor s. Weighting by $\phi_b(t-\ell)$, the inventory map $\mathbf{s}_k$, and $\widehat{c}_{kb}$ resolves the fitted contribution of each source group to each sensor by cell of origin. This is the puff response of Section 4 read backward from the sensor and requires no new transport operator. Per-sensor attribution uses only the rows of $\widetilde{H}_\Phi$ for that sensor: for the row submatrix $\widetilde{H}_\Phi^{(s)}$, $\widetilde{H}_\Phi^{(s)\top}\widetilde{H}_\Phi^{(s)} \preceq \widetilde{H}_\Phi^\top\widetilde{H}_\Phi$, so deleting rows cannot increase the smallest singular value or the rank and a single sensor is never more identifiable than the pooled network. Footprints and per-sensor contributions are therefore reported as spatial overlays of the global fit; per-sensor source *shares* are asserted only at the globally identifiable resolution, aggregating to a report group wherever the pooled diagnostics require one, which keeps the spatial view from implying separations the sensing system does not support.

L Uncertainty, Adequacy, and Reporting Details

This appendix expands the reporting components summarized in Section 6.3, and gives the IASA thresholds (Table 3).

Uncertainty reporting. Let $\widetilde{r} = \widetilde{Y} - \widetilde{H}_\Phi\widehat{\mathbf{c}}$ be the projected residual and $\widehat{\sigma}^2 = \|\widetilde{r}\|_2^2/\max(N - r_{\text{eff}}(\tau_\sigma), 1)$ a residual variance estimate. For an unconstrained or active-set approximation, let $\mathcal{A}_c = \{j : \widehat{c}_j > \tau_c\}$ and let $\widetilde{H}_{\Phi,\mathcal{A}_c}$ contain the active columns. The covariance of the active estimates is approximated by

$$\mathrm{Cov}(\widehat{\mathbf{c}}_{\mathcal{A}_c}) = \widehat{\sigma}^2(\widetilde{H}_{\Phi,\mathcal{A}_c}^\top\widetilde{H}_{\Phi,\mathcal{A}_c} + \lambda I)^\dagger. \tag{22}$$

The use of $N - r_{\text{eff}}(\tau_\sigma)$ is a heuristic degrees-of-freedom count that does not exactly account for the background rank r or the NNLS active-set constraints. Equation 22 is the standard active-set / ridge least-squares covariance approximation (Seber and Lee 2003): its diagonal entries give per-coefficient uncertainty and its off-diagonal entries expose coupled (jointly uncertain) estimates. For each transport ensemble member from Section 4, IASA rebuilds $\widetilde{H}_\Phi^{(b)}$, refits $\widehat{\mathbf{c}}^{(b)}$, and reports empirical quantiles across members as transport uncertainty.

Every ensemble is tagged by provenance: wind and dispersion members have `ensemble_kind=transport` and may form transport-uncertainty intervals, whereas alternative inventory maps, source locations, labels, or scales have `ensemble_kind=inventory` and produce scenario-wise robustness tables. The two kinds are never pooled into one interval, and active-set intervals describe the fitted observation model without converting inventory scenarios into probabilistic uncertainty.

Residual model-adequacy check. Let Z be an orthonormal basis for the background-orthogonal observed-row space, and define the nondegenerate residual coordinates and externally supplied noise covariance

$$\bar{r} = Z^\top(Y - H_\Phi^{\text{lag}}\widehat{\mathbf{c}}), \qquad \bar{\Sigma}_e = Z^\top\Sigma_e Z, \tag{23}$$

where Σ_e is an externally declared observation-noise covariance and Z removes the background directions so that $\bar{r}$ carries no degrees of freedom absorbed by Q. For independent homoscedastic noise $\Sigma_e = \sigma_e^2 I$, orthonormality of Z gives $\bar{\Sigma}_e = \sigma_e^2 I$ and $T_{\text{res}} = \|\bar{r}\|_2^2/\sigma_e^2$. When $\bar{\Sigma}_e$ is calibrated independently of this fitted residual and is positive definite, the adequacy statistic is

$$T_{\text{res}} = \bar{r}^\top\bar{\Sigma}_e^{-1}\bar{r}. \tag{24}$$

Its null distribution is not a plain χ^2, because $\widehat{\mathbf{c}}$ is fitted from the same data; IASA therefore calibrates it by parametric bootstrap. It simulates $B_{\text{res}} = 1000$ synthetic observation sets from the fitted source/background model plus the declared noise model, refits the coefficients on each draw, and recomputes T_{res} to form the null distribution. At $\alpha = 0.05$ it flags inadequacy when the observed statistic exceeds the empirical $(1-\alpha)$-quantile; the Monte Carlo p-value uses the standard add-one correction. If no externally calibrated noise model is supplied, IASA reports raw and projected residual norms and sensor-wise, time-wise, and autocorrelation summaries with `calibration_status=uncalibrated` and emits no pass/fail adequacy claim. Rejection establishes a mismatch with the declared model but does not identify a missing source, and non-rejection does not establish inventory completeness: an omitted source whose sensor-time signature lies in $\mathrm{span}[H_\Phi^{\text{lag}}, Q]$ can be absorbed by fitted terms and remain residual-invisible.

Wind-distribution diagnostics. Diagnostics for one realized wind window answer a retrospective question. Prospective network adequacy is assessed empirically over predeclared historical or simulated wind windows. Across those response matrices, IASA reports the 5th, 50th, and 95th percentiles of σ_J, the probability of full numerical and effective rank, the probability that each coefficient is weak, the probability that each source pair is ambiguous, and the frequency of each conservative report component. These are Monte Carlo or empirical distributional summaries under the sampled wind population, not a new identifiability theorem.

Acceptance criteria for refinement. If the optional constrained refinement of Appendix I is used, it is accepted only if it improves fit without degrading separability. Let $\widetilde{H}_{\Phi,0}$

Table 4: New Delhi source groups, spatial proxies, and temporal-basis components.

Source group	Spatial proxy	Temporal-basis components
Brick kilns	regional kiln map	alternating 12-hour blocks
Industries	regional industry map	continuous + daytime (07–19)
Population density	gridded population	peaks 07:00, 13:00, 19:00
Traffic	road-network map	diurnal slots 00/06/12/18

be the projected response before refinement and $\widetilde{H}_{\Phi,\text{ref}}$ after refinement. We require $\sigma_J(\widetilde{H}_{\Phi,\text{ref}}) \geq (1-\eta_{\text{id}})\sigma_J(\widetilde{H}_{\Phi,0})$ and $\max_{i\neq j,\, i,j\notin\mathcal{W}} \rho_{ij}^{\text{ref}} \leq \tau_\rho^{\text{ref}}$. These checks prevent end-to-end tuning from improving sensor fit by making source–basis fingerprints less distinguishable. If either response has fewer than J numerically nonzero singular values, its σ_J is taken to be zero (Section 5.3), so refinement cannot be accepted by exploiting a rank-deficient response.

Reporting protocol. The source ambiguity graph, its deterministic connected components, and the transitive overmerging of an A–B–C chain are defined in Section 5.4. Operationally, those components are the conservative recommended report groups, each edge retains its triggering coefficient pair, and rank deficiency without an eligible pair yields a global unresolved warning rather than an invented edge. For each reported source group G_a, IASA reports the activity trajectory $\widehat{\boldsymbol{\theta}}_{G_a,1:T}$, its interval $\text{CI}(\widehat{\theta}_{G_a})$, member visibilities $\{v_{kb} : k \in G_a\}$, and the maximum eligible cross-group coherence $\max_{k\in G_a,\, k'\notin G_a,\, (k,b),(k',b')\notin\mathcal{W}} \rho_{(k,b),(k',b')}$. A source contribution is marked reliable only when its constituent reported coefficients are above the visibility threshold, its uncertainty interval is sufficiently narrow, and its eligible coherence with other reported groups is below the ambiguity threshold. Reports retain the basis names of every weak coefficient and the coefficient pair attaining each cross-source coherence maximum; coherence and ray distance involving weak fingerprints are displayed as "N/A" and excluded from extrema, while the weak flags remain visible. Otherwise IASA labels the source as weakly visible, ambiguous with another source group, or merged into a coarser identifiable group.

M New Delhi Platform Details

This appendix expands the platform construction of Section 7.1: the source inventories (Table 4), the kriged initial-condition baseline, and the auxiliary simulator.

Kriged initial-condition baseline. For observed New Delhi runs, the initial pollution field is estimated from the first available regulatory observations (the two Pusa monitors averaged) by a normalized Gaussian-kernel spatial interpolation on the response grid, a kriging surrogate that avoids an additional heavyweight geostatistics dependency. Its zero-source transported trajectory, obtained by propagating this initial field as a single release through the same open-boundary puff operator (its Gaussian kernel is the advection–diffusion Green's function, so units are preserved), is subtracted from the sensor observations before fitting source activities. Source coefficients are therefore fitted to the residual sensor-time signal after removing the contribution of the initialized background field and the low-rank temporal background basis. Kernel interpolation from sparse first-hour observations is itself an ill-posed spatial estimate, so this baseline injects some uncertainty; the effect is limited by declaring and subtracting the baseline before fitting and by the low-rank background basis Q, which absorbs residual broad structure. Controlled operator-matched experiments may instead use a zero initial field when the synthetic data are generated from the same zero-initial-state response.

Auxiliary advection–diffusion simulator. The auxiliary advection–diffusion simulator uses first-order upwind advection, a five-point Laplacian, a two-stage Heun update, diffusivity 3×10^{-4}, and edge-hold boundaries. It provides a structural forward-model mismatch: data generated by this distinct transport operator are fit with the open-boundary puff response, testing the puff approximation against a different operator family. This mismatch also drives the residual-adequacy test of Section 6.3, letting us check its power against structural transport misspecification and not only against missing sources.

N Additional Experiments

This appendix reports the six controlled experiments deferred from Section 7: coherence and grouped reporting, transport error, inventory robustness, lag-window selection, temporal-basis recovery, and per-sensor footprints.

N.1 Coherence and Grouped Reporting

We form increasingly coherent source pairs by shifting a copy of one inventory map by a controlled spatial offset, so that decreasing the shift drives coherence toward one. Individual-source errors are compared with errors in the summed activity and fitted sensor contribution of each recommended connected component. Edge reports retain the triggering source–basis pair.

Table 5 reports the coherence sweep. Shrinking the spatial offset drives maximum eligible coherence from 0.05 to 0.96 while σ_J falls from 21.8 to 4.6. At every tested offset the conservative merge rule kept the two sources in separate singleton components and both were recovered exactly (individual and grouped relative error 0), so no grouped reporting was required and no cross-source trigger pair fired; the eligible ray distance and triggering pair are therefore undefined (null) rather than forced to a number. The reporting layer retains *every* trigger edge without deduplication when a merge does fire (an A–B–C chain keeps both edges); no merge fired in this experiment.

N.2 Transport Error

We use two kinds of transport error. *Parametric* perturbations vary wind speed, wind direction, and dispersion within the puff family; a *structural* mismatch instead generates observations

Table 5: Experiment 2: coherence rises as the source offset shrinks, but the conservative rule merged no sources and recovery stayed exact.

offset	max. coh.	σ_J	merged	indiv./grp. err.
8.0	0.048	21.77	no	0.0 / 0.0
6.0	0.862	10.48	no	0.0 / 0.0
4.0	0.905	8.40	no	0.0 / 0.0
2.0	0.945	5.43	no	0.0 / 0.0
1.0	0.958	4.64	no	0.0 / 0.0

Table 6: Experiment 5: parametric transport error. Coefficient error rises with the operator error norm on the direction and dispersion axes (the speed axis is non-monotonic). Structural PDE mismatch: adequacy rejection rate 1.0.

axis	value	op. err.	σ_J	coef. err.
direction (°)	0	0.00	0.87	0.00
direction (°)	5	0.29	3.37	0.62
direction (°)	10	0.60	7.07	0.80
direction (°)	20	0.87	7.98	0.86
speed (×)	1.25	0.29	2.09	0.45
speed (×)	1.50	0.53	4.57	0.27
dispersion (×)	1.50	0.19	1.42	0.50
dispersion (×)	2.00	0.35	1.76	0.89

with the auxiliary edge-hold advection–diffusion PDE while fitting with the open-boundary puff response. The operator error norm, coefficient/activity error, residual, and singular spectrum are reported together, and the parametric transport ensembles may support transport-uncertainty intervals. The structural case additionally exercises the residual-adequacy test of Section 6.3: we report its rejection rate to show it has power against forward-model misspecification, not only against missing sources.

Table 6 reports the parametric family: coefficient error grows with the operator error norm along the wind-direction (to 0.86 at $20°$) and dispersion (to 0.89 at $2\times$) axes, confirming that transport error propagates into attribution rather than being absorbed. The wind-speed axis is non-monotonic at these magnitudes (error 0.45 then 0.27 as the speed factor rises from 1.25 to 1.5), a reminder that a coarse operator-error norm does not fully order attribution error. The structural PDE mismatch (operator mismatch norm 0.44) is the sharper test: it drives the mean coefficient relative error to 0.86, and the refitted-bootstrap adequacy test rejects in 100% of trials, confirming its power against forward-model misspecification. (Transport-interval coverage is supported by the parametric ensemble but not tabulated per row here.)

N.3 Inventory Robustness

We perturb inventory locations, spatial scales, category assignments, and map versions while holding transport fixed. Each alternative inventory is a separate robustness scenario. Its variation is not pooled with the transport error of Experiment 5 and is not reported as a confidence interval.

Table 7 reports the four scenarios (location shift, spatial rescale, alternative map version, category swap) on the real New Delhi brick-kiln and industry maps. The known coefficients $[1.0, 0.6]$ are recovered exactly in every scenario, but the identifiability geometry moves with the inventory version: rescaling the map or substituting an alternative version nearly doubles σ_J (to 47.0 and 46.0) relative to baseline (21.6), whereas a symmetric category swap leaves it unchanged. The attribution *target* thus changes with the assumed inventory even when recovery is exact.

Table 7: Experiment 6: inventory robustness on the real maps. Recovery is exact in every scenario; σ_J tracks the inventory version.

scenario	σ_J	coef. err.
baseline	21.56	0.0
location shift	28.85	0.0
spatial rescale	47.00	0.0
alt. map version	46.01	0.0
category swap	21.56	0.0

Table 8: Experiment 7: lag-window selection. Conditioning improves with lag; the selected lag ($L = 16$) is chosen from geometry alone, not fitted coefficients.

lag L	σ_J	κ	#comp.	coef. err.
4	0.08	569.2	2	0.0
6	0.71	63.8	2	0.0
8	2.46	18.6	2	0.0
10	4.57	10.1	2	0.0
12	6.16	7.5	2	0.0
16	9.07	5.1	2	0.0

N.4 Lag-Window Selection

Physical travel and residence times define the candidate grid. We evaluate Equation 1, choose the smallest lag satisfying $\tau_L = 10^{-3}$, and report rank, conditioning, and report-component stability across the grid without using fitted source coefficients for selection.

Table 8 sweeps the physically-motivated candidate grid. As the lag window grows the response stabilizes and conditioning improves monotonically (κ falls from 569 at L=4 to 5.1 at L=16; σ_J rises from 0.08 to 9.07), while the numerical rank (2) and the number of report components (2) stay fixed and recovery is exact throughout. The selection rule picks the smallest lag whose relative Frobenius change falls to $\tau_L = 10^{-3}$ (here $L = 16$), using only response-matrix geometry; fitted source coefficients are not consulted and the row count is invariant across the grid.

N.5 Temporal-Basis Recovery

Known traffic diurnal, brick-kiln intermittent, industry day/night, and mixed source-basis coefficients are reconstructed at increasing noise. This study separates coefficient error from error in the reconstructed source activity trajectory.

Table 9 reports temporal-basis recovery over the diurnal, intermittent-block, and day/night bases at increasing noise

Table 9: Experiment 9: temporal-basis recovery (bases: diurnal, block, day/night). Activity-trajectory error stays below coefficient error at every noise level.

noise	coef. err.	activity err.
0.00	0.000	0.000
0.02	0.132	0.075
0.05	0.329	0.188
0.10	0.659	0.375
0.20	0.900	0.582

(fixed $\sigma_J = 0.14$). Separating the two error notions is the point: the reconstructed source-activity trajectory error is consistently smaller than the raw coefficient error (e.g. 0.58 versus 0.90 at 20% noise), because projecting the noisy coefficients through the temporal bases averages out part of the coefficient noise in the activity domain. The activity-to-coefficient error ratio is near-constant (≈ 0.57) across noise levels, indicating that this damping is a fixed geometric property of the temporal-basis map (which amplifies the coherent true signal more than the isotropic estimation noise) rather than a noise-dependent effect.

N.6 Per-Sensor Footprints

On controlled trials with known source origins, we check that the per-sensor footprints of Section 5.4 localize the responsible upwind cells and that per-sensor contributions sum to the fitted sensor signal. On observed New Delhi data we report, per monitor, the fitted source-group contributions and their spatial cells of origin, aggregated to the identifiable report groups so no per-sensor separation exceeds the network's global resolution.

On the controlled trial this localization is accurate: the footprint mass centroid lies 0.74 cells from the true source origin, 100% of the footprint mass falls within the 4-cell localization radius, and the footprints are nonnegative. The exact per-sensor decomposition holds: per-sensor contributions sum to the fitted sensor signal to machine precision (contribution-sum error 0), and because a non-singleton report component is present, footprints are reported at the group granularity. The observed New Delhi per-monitor footprints appear with the apportionment results in Section 7.4.